Using VR and Objective Approach for Assessing Psychiatric Variables

Thesis

Submitted in partial fulfillment of the requirement for the

Master's Degree in Data Science and Engineering

at

The City College of the City University of New York

by

Safal Thapa

Approved by:



Professor Dongrong Xu



Professor Zhigang Zhu

Thesis Advisors



Professor Akira Kawaguchi

Department Chair, Computer Science

May 2026

Using VR and Objective Approach for Assessing Psychiatric Variables

Data Science and Engineering Program

The City College of the City University of New York



Safal Thapa

Mentors: Professor Dongrong Xu & Professor Zhigang Zhu

May 2026

Acknowledgements

I extend my deepest gratitude to my advisor, Prof. Dongrong Xu and Prof. Zhigang Zhu, for the opportunity to work under their guidance. Their constant feedback and guidance kept the project on track and made every hour of research meaningful. I am truly grateful for their patience, their belief in my work and the time they generously dedicated to helping me grow as a researcher.

I would also like to thank the faculty and staff of the Data Science and Engineering Department for equipping me with the knowledge and skills I needed throughout my graduate studies. The coursework and learning experience provided during this program laid a strong foundation for this thesis and shaped the way I think about the research.

This journey was not always easy or straightforward, especially since this was an entirely new domain for me. However, the support I received along the way made it possible. I am deeply grateful to everyone who played a part, however big or small, in helping me reach this point.

Table of Contents

ABSTRACT **1**

CHAPTER 1: INTRODUCTION **4**

1.1 BACKGROUND 4

1.2 RESEARCH PROBLEM 5

1.3 MOTIVATION OF THE RESEARCH 6

1.4 OBJECTIVES 7

1.5 EXPECTED FINDINGS 7

1.6 CONTRIBUTIONS OF THIS RESEARCH 8

1.7 OUTLINE OF THE THESIS 8

CHAPTER 2: RELATED WORK **9**

2.1 OVERVIEW 9

2.2 MACHINE LEARNING AND GRAPH METHODS FOR MDD RESEARCH 9

2.3 VIRTUAL REALITY AS A BEHAVIORAL ASSESSMENT TOOL 10

2.4 VR PARADIGMS FOR RESPONSE INHIBITION ASSESSMENT 10

2.5 MULTIDIMENSIONAL FRAMEWORK FOR IMPULSIVITY ASSESSMENT 11

2.5 SUMMARY AND RESEARCH GAP 12

CHAPTER 3: DATA **14**

3.1 PARTICIPANTS 14

3.2 CLINICAL ASSESSMENT 15

3.3 VR SHOOTING TASK 16

3.4 DTI ACQUISITION 17

CHAPTER 4: METHOD **18**

4.1 CONNECTOME CONSTRUCTION 18

4.2 GRAPH METRIC COMPUTATION 18

4.2.1 Confirmatory Global Metrics *18*

4.2.2 Expanded Nodal Metrics *20*

4.2.3 Modular and Summary Metrics *20*

4.2.4 JHU White Matter Tract Metrics *21*

4.3 VR Feature Extraction and PCA-Based Features 22
4.4 Pre-Shot Time Window Analysis 22
4.5 Statistical Methods 23
4.5.1 Confirmatory Regression *23*
4.5.2 Sensitivity Analyses *24*
4.5.3 Nodal Regressions *24*
4.5.4 Multivariate and Machine Learning Methods *24*
4.5.5 Power and Sensitivity Analysis *25*
4.5.6 Age Sensitivity Analysis *25*
4.5.7 JHU White Matter Tract Correlations *26*
CHAPTER 5: RESULTS 27
5.1 Confirmatory Analysis: Brain Metrics and VR Outcomes 27
5.1.1 Effect Size Analysis *28*
5.2 Sensitivity Analyses 29
5.2.1 Cook's Distance — Influential Observation Analysis *30*
5.3 Brain, VR and Clinical Associations 31
5.3.1 BIS-11 and Clinical Scale Associations *31*
5.3.2 Brain Metrics and Clinical Scale Associations *31*
5.3.3 VR Outcomes and Clinical Scale Associations *32*
5.4 VR Impulsivity Composites and Validation 32
5.5 Pre-Shot Time Window Results 34
5.5.1 Window Profile Results *34*
5.5.2 SSI-19 Window Profile — Validation of the 0.6 Second Window *35*
5.5.3 Feature-Specific Optimal Windows *36*
5.5.4 Within-Subject and Between-Subject Kinematic Results *37*
5.6 Expanded Network Analysis — Hub Structure 38
5.7 Modular Network Metrics and Clinical Results 39
5.8 Nodal Regression Results — 120 AAL2 Nodes 41
5.9 Multivariate and Machine Learning Analysis Results 43
5.10 Age Sensitivity Analysis 45
5.10.1 Main Sensitivity: Age as Covariate *45*

5.10.2 Supplementary Age Analyses *46*
5.11 White Matter Tract Analysis (JHU Atlas) 47

CHAPTER 6: DISCUSSION 49

6.1 Overview of Findings 49
6.2 Frontal Hub Connectivity and Motor Control 50
6.3 Temporal Pole and Cerebellar Connectivity 51
6.4 Hub Organization and Trait Impulsivity 52
6.5 Impulsivity at Different Time Scales in Pre-Shot Behavior 53
6.6 Time Window Analysis and Implications for VR Tasks 54
6.7 Possible Links Between Brain, VR Behavior, and Clinical Symptoms 55
6.7.1 VR Cognitive Impulsivity and Suicidal Ideation *55*
6.7.2 Brain, VR, and Clinical Relationships *55*
6.8 Limitations 56
6.9 Age Effects and Brain-Impulsivity Relationship in MDD 58

CHAPTER 7: CONCLUSIONS AND FUTURE DIRECTIONS 60

7.1 Summary 60
7.2 Key Contributions 60
7.3 Future Directions 61

REFERENCES 63

APPENDIX A: FULL FIGURE GALLERY 68

Original Confirmatory Figures 68
Figure A1. Study Design Overview. 68
Figure A2. Clinical Characteristics. 69
Figure A3. Small-Worldness Analysis. 70
Figure A4. VR Composites Detailed 70
Figure A5. Pre-Shot Correlation Summary. 71
Extended Analysis Figures 71
Figure A7. Window Optimization Profiles *71*
Figure A8. Window × Outcome Heatmap *72*

Figure A9. Hub Architecture *72*
Figure A10. New Metrics × Clinical Heatmap *73*

APPENDIX B: SUPPLEMENTARY TABLES 73

B1 Top 24 Hub Nodes 73
B2 Bootstrap Window Optimization Results 74
B3 Outcome-Specific Optimal Windows 75

Abstract

Background: Major Depressive Disorder (MDD) is associated with several clinical features, including impulsivity, anxiety, depression severity, and suicidal ideation, which are all related to difficulties in behavioral control. Because of this, it is important to understand what is happening in the brain behind impulsive behavior. One way to study this is using diffusion tensor imaging (DTI) and network analysis to look at brain structural connectivity. However, it is still not very clear how whole-brain network organization relates to real-world impulsive behavior in MDD, including at the level of individual regions, network structure, and changes over time.

Objectives: In this thesis, we focused on four main questions. First, whether structural brain network measures from DTI can predict impulsive behavior in a VR shooting task in patients with MDD. Second, whether more detailed network measures, such as nodal betweenness, hub disruption, rich-club coefficient, within-module z-score, and participation coefficient, can explain additional information beyond the global metrics. Third, whether pre-shot motor features are related to clinical severity. And finally, which pre-shot time window works best for predicting clinical outcomes.

Methods: We started with 181 preprocessed diffusion-weighted neuroimaging datasets from clinical recruitment. For participants with duplicate sessions, we retained the session with the lowest head motion, measured by the eddy mean relative RMS displacement.

Participants were included in the final sample if they met all of the following criteria: (i) a confirmed MDD diagnosis; (ii) complete DWI data with a successfully reconstructed structural connectome; and (iii) complete VR task data. The final sample included 78 MDD patients (59 females, 19 males; age range 12–38 years, mean = 19.8, SD (standard deviation) = 6.4; mean HDRS = 17 (Hamilton Depression Rating Scale -17 items), indicating severe depression; mean BIS-11 (Barratt Impulsiveness Scale, Edition 11) score = 94.3, above the high-impulsivity threshold of 72).

For the brain network analysis, we included four global topology measures: characteristic path length, clustering coefficient, global efficiency, and modularity. We also included betweenness centrality for three pre-specified hub regions: the right inferior frontal gyrus, supplementary motor area, and anterior cingulate cortex. VR behavioral performance was summarized using three PCA-derived components: Motor (D1), Cognitive (D2), and Non-planning (D3). In addition, we also included the raw collimation timer scores. The clinical measures included BIS-11, SSI-19, HAMD-17, HAMA, BAI, STAI, and SHAPS.

For the confirmatory analysis, we used OLS regression while controlling for gender and head motion, followed by Benjamini–Hochberg FDR correction. The expanded analysis included nodal regression across 2,520 models, based on 120 nodes, 3 metrics, and 7 outcomes, which included two collimation timer thresholds and five clinical scales. We also examined modular measures such as within-module Z-score and participation coefficient. In addition, we performed partial least squares, ridge regression with leave-one-out cross-validation to predict SSI-19, and a pre-shot kinematic window analysis across seven-time windows from 0.2 to 0.8 seconds. Finally, we performed age sensitivity analyses by comparing adolescents and adults and by adding age as an additional covariate.

Results: Three brain–VR associations remained significant after FDR correction. Right inferior frontal gyrus betweenness centrality showed a positive association with collimation timer performance ($\beta = 2.72$, $q = 0.002$, $\Delta R^2 = 21.5\%$). Characteristic path length and clustering coefficient were also significantly related to collimation timer outcomes ($q < 0.01$). These effects remained stable in age adjusted analyses. Path length and clustering effects were stronger in adolescent patients ($n = 41$) compared to adults ($n = 37$). The frontal betweenness effect was consistent across both age groups.

In the expanded nodal analysis, we found 81 significant associations. The strongest effects came from left temporal pole local efficiency and cerebellar degree, both related

to collimation timer performance. Within-module Z-score was also related to BIS-11 total ($\rho = +0.34$, $p = 0.002$) and motor impulsivity ($\rho = +0.32$, $p = 0.004$). We also found that a VR-based cognitive impulsivity composite was positively associated with suicidal ideation ($\rho = +0.37$, $q = 0.024$).

Partial least squares showed a significant brain–behavior relationship ($r = 0.459$, $p < 0.001$). Ridge regression explained about 24% of the variance in suicidal ideation. The pre-shot window analysis showed a fairly stable pattern across 0.2–0.8 seconds, which supports using the 0.6-second reference window.

Conclusions: The results suggest that physical brain organization in terms of structural network may be related to both impulsive behavior and clinical severity in MDD. Some key areas that came up include frontoparietal hub connectivity, cerebellar-frontal white matter, temporal pole local efficiency, and within-module connectivity. These associations were still present after we controlled for age, which suggests they are not just due to age differences in our mixed adolescent–adult sample. At the same time, path length and clustering effects looked stronger in adolescent patients, which may point to some developmental differences. This is something that would need to be explored more carefully in future studies with age-matched groups.

The VR task also showed some meaningful patterns. It was consistently aligned with BIS-11 scores, the finding supports the predictive validity for suicidal ideation and elucidate patients respond under stress. The 0.6-second pre-shot window seemed to work best for predicting suicidal ideation. Overall, these results suggest that combining brain network analysis with VR-based behavioral measures may be useful as an additional tool alongside standard clinical assessments, but this would need more validation.

Keywords: Major Depressive Disorder, DTI Connectomics, Graph Theory, AAL2 Parcellation, Virtual Reality, Impulsivity, BIS-11, Suicide Ideation, Small-Worldness, Temporal Pole, Cerebellum, Pre-Shot Window Analysis, Hub Disruption, Adolescent MDD

Chapter 1: Introduction

1.1 Background

Major Depressive Disorder (MDD) is one of the leading causes of disability worldwide, affecting over 280 million people globally [1]. Besides the well-known symptoms like persistent low mood and loss of interest, MDD is also associated with a range of cognitive and behavioral difficulties. One of the most clinically important of these is impulsivity.

Impulsivity in MDD is not a single construct but has several dimensions. These include acting without thinking (motor impulsivity), difficulty planning ahead (non-planning impulsivity), and making fast but inaccurate decisions under uncertainty (cognitive impulsivity) [2]. These dimensions are captured by the Barratt Impulsiveness Scale (BIS-11), which is the most widely used tool for measuring impulsivity [3]. Importantly, impulsivity in MDD is linked to serious outcomes including treatment non-adherence, self-harm, and suicidal behavior [4]. Studies have also shown that impulsivity scores are higher in MDD patients compared to healthy individuals. Within MDD patients, higher impulsivity is associated with greater depression severity and suicidal ideation [4].

Impulsivity is usually measured using self-report questionnaires like the BIS-11. However, these methods have limitations. People may not always accurately report how they actually behave, and their answers can be affected by their mood at the time of testing [5]. This means questionnaires alone may not give a complete picture of impulsivity in real behavior. Virtual reality (VR) can help with this. VR creates environments that feel more like real life, which makes it possible to observe how a person actually behaves in a controlled setting [6]. In this study, we used a VR shooting task where participants had to shoot targets while dealing with wind disturbances that continuously shifted the bullet path. To shoot accurately, participants had to observe the wind, adjust their aim, and wait for the right moment before firing. Participants who fired too quickly, without properly adjusting, showed their impulsive behavior through their actions rather than through self-report.

Brain structure is also important when looking at impulsive behavior. White matter pathways connect different parts of the brain and allow them to communicate with each other. In MDD, some of these pathways can be disrupted, especially those linking the prefrontal cortex with regions involved in emotional processing [7]. The prefrontal cortex is known to play a role in planning and self-control, so changes in these connections may affect behavior. These connections can be studied using diffusion tensor imaging (DTI). When we combine DTI with graph theory analysis, we can look at how the brain is organized as a network [8]. This includes things like how efficiently information moves across the brain, how strongly nearby regions are connected, and how important certain regions are within the overall network [9].

However, it is still not fully clear how whole-brain structural connectivity relates to impulsive behavior in people with MDD, especially when behavior is measured in realistic settings [10]. In this thesis, we explore that relationship by combining DTI-based network analysis with behavioral data from a VR shooting task.

1.2 Research Problem

Psychiatric disorders like MDD are often difficult to diagnose. One reason is that their symptoms are not very specific. They can overlap with other conditions and also look different from one person to another [1]. Right now, diagnosis mostly depends on clinical interviews and questionnaires. These methods rely a lot on patients' willingness to report as well as their awareness of the metal status, and also heavily on the clinician's judgment, therefore may not always capture important information on symptoms like impulsivity [5].

This becomes a problem because impulsivity in MDD is linked to higher risk of self-harm and suicide [4]. If we are not able to measure it clearly, it becomes harder to assess risk and provide the right treatment.

There is also a gap in current research. Many studies focus on global brain network measures, but it is still not very clear whether more detailed measures, like node-level and modular metrics, can give additional information about impulsive behavior [9, 10]. Because of this, there is a need for more objective approaches that can connect brain structure, real-world behavior, and clinical assessment in a more reliable way.

To address these issues, this thesis focuses on the following research questions:

1. Does DTI-based structural network topology predict symptomatic behaviors (such as impulsivity, anxiety, depression, suicidal ideation) measured using a VR shooting task in MDD patients?
2. Do more detailed network measures, such as nodal betweenness, hub disruption, rich-club coefficient, within-module Z-score, and participation coefficient, provide additional information beyond global metrics?
3. Are pre-shot motor features related to clinical severity in MDD patients?
4. What is the most suitable pre-shot time window for predicting clinical outcomes?

1.3 Motivation of the Research

Diagnosing psychiatric disorders involves a temporal requirement with an intention to differentiate acute stress reactions from chronic clinical conditions. For example, schizophreniform disorder requires consistent symptoms for at least 1 month whereas schizophrenia needs at least 6 months, and MDD requires a minimal time period of 2 weeks (DSM-V), which means treatment may also be delayed [1]. Since these conditions affect both brain connectivity and behavior [7, 8], combining VR-based assessments, brain imaging, and machine learning could support diagnosis in a more objective way.

Understanding the neural basis of impulsivity is especially important because it can lead to dangerous outcomes, including suicide attempts [4]. Currently, most assessment tools rely on self-report, which has clear limitations when it comes to capturing real behavior [5]. VR offers a different approach. It allows us to recreate realistic and mildly stressful situations in a safe environment and observe how people actually respond. This is something paper-based tests simply cannot do [6].

One useful feature of the VR task is the short period just before a participant pulls the trigger — called the pre-shot window. The movement features captured during this time, such as hand speed, acceleration, and head scanning behavior, can reflect how the decision-making process is actually unfolding. This kind of detailed behavioral data, combined with brain imaging, could help identify high-risk patients earlier and support better clinical decisions [8, 9].

1.4 Objectives

The main goal of this thesis is to understand the relationship between structural brain connectivity and impulsive behavior in MDD patients. To do this, we set the following objectives:

1. To examine whether DTI-derived structural network topology can predict impulsive behavior measured through a VR shooting task in MDD patients.
2. To explore whether expanded network metrics — including nodal betweenness, hub disruption, rich-club coefficient, within-module Z-score, and participation coefficient — can capture clinical and behavioral variance beyond what global metrics alone provide.
3. To investigate whether pre-shot motor features are associated with clinical severity in MDD patients.
4. To identify the best pre-shot time window for clinical prediction.

1.5 Expected Findings

Based on previous research, we expected that MDD patients with more disrupted hub connectivity and reduced small-worldness would show more impulsive behavior in the VR task [8, 9]. We also expected that expanded nodal metrics — particularly those related to frontal and cingulate regions — would reveal brain-behavior relationships that global metrics alone would not capture [9]. For clinical measures, we expected VR-derived behavioral features to show meaningful agreement with BIS-11 scores [3] and to predict suicidal ideation severity to some degree [4]. Finally, we expected that pre-shot

motor features would reflect differences in clinical severity, with more impulsive patients showing faster and less controlled movements before firing [6].

1.6 Contributions of this Research

This thesis makes the following contributions to the field:

1. This study is one of the first to directly connect DTI-based whole-brain structural network topology with ecologically valid impulsive behavior, measured through a VR task in MDD patients [8, 10].
2. It goes beyond commonly used global brain metrics by testing expanded nodal metrics — including nodal betweenness, hub disruption, rich-club coefficient, within-module Z-score, and participation coefficient — giving a more detailed picture of how brain network organization relates to impulsive behavior [9].
3. It shows that a VR shooting task can capture meaningful behavioral differences in MDD patients, with good agreement with standard clinical measures like BIS-11 and the ability to predict suicidal ideation [3, 4, 6].
4. It shows that the 0.6-second pre-shot window is the most reliable time window for predicting clinical outcomes, which can help guide how future VR-based assessments are designed [6].

1.7 Outline of the Thesis

This thesis is organized as follows. Chapter 1 provides the introduction, including the background, research problem, motivation, objectives, and contributions of this work. Chapter 2 reviews related work on brain connectivity, impulsivity in MDD, DTI-based connectomics, and the use of VR in psychiatric assessment. Chapter 3 describes the data used in this study, including the DTI brain imaging data and the VR shooting task. Chapter 4 explains the methodology, covering brain network construction, graph theory analysis, and the statistical models used. Chapter 5 presents the results and discusses them in relation to the research questions. Chapter 6 discusses the findings in relation to prior research and the study's limitations. Chapter 7 concludes the thesis with a summary of the key findings, contributions, and directions for future research.

Chapter 2: Related Work

2.1 Overview

This chapter looks at previous studies that are relevant to this thesis. The literature we reviewed covers two main areas: machine learning and graph-based methods for MDD detection using brain imaging, and the use of virtual reality as a behavioral assessment tool. The chapter ends with a summary of the main gaps in the literature that this thesis tries to address.

2.2 Machine Learning and Graph Methods for MDD Research

Several studies have explored how machine learning can be used to detect and classify MDD from brain imaging data. Qin et al. combined structural and functional MRI using deep learning and showed that using both data types together improved classification compared to using either alone [22]. This supports the value of combining brain connectivity data with behavioral measures, which is the approach taken in this thesis.

A review by Librenza-Garcia et al. found that brain connectivity features are useful for distinguishing MDD patients from controls, and that combining imaging data with clinical information generally improves model performance [23]. Jiang et al. built on this by showing that deep learning models trained on functional connectivity data could predict MDD onset up to two years in advance, with key predictive regions overlapping with areas involved in impulsivity [24]. They also found that model performance declined in older adults — a relevant observation given that our sample spans a wide age range.

Zhao et al. applied ensemble graph neural networks to resting-state fMRI data for MDD detection, achieving 75.8% accuracy and highlighting common practical challenges like class imbalance [25]. While this thesis uses structural rather than functional connectivity, the graph-based representation of brain networks is consistent with the graph-theoretic approach used here.

2.3 Virtual Reality as a Behavioral Assessment Tool

A growing number of studies have used VR to collect behavioral data for psychiatric assessment. This approach has clear advantages over self-report measures because it captures how people actually behave in realistic situations.

Huang et al. developed a VR-based framework for screening depression in adolescents, collecting EEG, eye-tracking, and heart rate data during a VR task [26]. Their model achieved 81.7% accuracy, showing that VR-based behavioral data can be a reliable and objective measure of psychiatric symptoms. Pollak et al. used a VR performance task to assess ADHD in adults and found that movement features — such as head speed and total distance traveled — were among the strongest predictors of hyperactivity and inattention [27]. This is especially relevant to this thesis because it shows that motor features captured during a VR task can carry meaningful clinical information. In this study, similar features — including hand acceleration, speed variability, and head scanning behavior — are extracted from the pre-shot window and tested for their relationship with impulsivity and clinical severity.

2.4 VR Paradigms for Response Inhibition Assessment

Response inhibition refers to the ability to suppress or stop a planned motor response and is widely considered one of the core behavioral measures of impulsivity. It is most commonly evaluated using Stop-Signal and Go/No-Go paradigms, which are well established for assessing inhibitory control [41]. More recently, these paradigms have been adapted into VR environments, where inhibitory tasks are embedded within more naturalistic and interactive settings [42]. Compared to traditional computer-based tasks, VR allows researchers to collect additional behavioral information such as movement trajectories, gaze patterns, and body kinematics, while also improving ecological validity.

Several studies have shown that VR-based response inhibition tasks produce findings consistent with traditional laboratory paradigms. Ajana et al., for example, developed an immersive VR visual search task that evaluated both spatial and distractor inhibition

and successfully reproduced established behavioral effects using response time and movement-based measures [43]. Other studies have extended this approach by integrating VR paradigms with neuroimaging techniques. Dina et al. combined functional near-infrared spectroscopy with a naturalistic VR inhibitory control task and reported significant associations between task performance and prefrontal cortical activity in children [44].

These findings are highly relevant to the current thesis. In the VR shooting paradigm used in this study, participants were required to suppress impulsive firing responses and intentionally delay trigger execution before taking a shot. This behavioral process closely aligns with the fundamental mechanism underlying response inhibition. Importantly, the right inferior frontal gyrus identified in the present analysis has consistently been reported in prior response inhibition literature as a key component of the brain's inhibitory control network [41].

2.5 Multidimensional Framework for Impulsivity Assessment

Impulsivity is widely recognized as a multidimensional construct rather than a single behavioral trait. The different components of impulsivity, including response inhibition, cognitive control, delay discounting, and risk-taking behavior, reflect distinct underlying processes and together provide a broader understanding of impulsive behavior across both clinical and non-clinical populations [45].

Among these dimensions, cognitive control plays a major role in regulating impulsive behavior. Cognitive control includes processes such as working memory, attentional regulation, and goal-directed decision-making, all of which have been consistently associated with impulsivity levels across psychiatric conditions, including MDD [46]. Deficits in these functions are often linked to impulsive decision-making and impaired behavioral regulation. This framework is particularly relevant to the current thesis because the VR-derived behavioral composites obtained through PCA align closely with these cognitive dimensions. Specifically, the Cognitive composite (D2) reflects deliberate

decision-making and behavioral regulation before each shot, while the Non-planning composite (D3) captures more reckless and poorly regulated behavior during challenging shooting conditions. These dimensions closely correspond to the cognitive and non-planning components measured by the BIS-11 subscales.

This multidimensional perspective also supports the analytical approach used in this study. Instead of treating impulsivity as a single outcome measure, the present work evaluated relationships between brain network metrics and multiple behavioral and clinical dimensions simultaneously, including motor, cognitive, and non-planning components. This approach is consistent with current theoretical and clinical models of impulsivity, which emphasize that impulsive behavior emerges from several interacting cognitive and behavioral processes rather than a single unified mechanism [45, 46].

2.5 Summary and Research Gap

The studies reviewed throughout this chapter demonstrate that machine learning and neuroimaging approaches can provide valuable insight into MDD, while VR-based behavioral paradigms can capture clinically meaningful behavioral information that is often difficult to obtain through traditional self-report measures alone. Prior work has also consistently shown that integrating brain imaging, behavioral measurements, and clinical assessments generally produces stronger and more reliable findings than relying on a single modality in isolation [28]. In addition, recent research on VR-based response inhibition and multidimensional impulsivity assessment suggests that motor behavior, movement patterns, and task-related kinematic features contain meaningful clinical information that can reflect underlying cognitive and behavioral processes.

Despite these advances, several important gaps remain in the literature. Most existing machine learning studies in MDD have primarily focused on functional MRI data, while relatively fewer studies have examined DTI-based structural connectomes for predicting behavioral measures related to impulsivity. Similarly, although global brain network properties have been widely investigated in MDD, expanded nodal-level metrics such as within-module degree z-score, participation coefficient, and hub disruption index have

not been extensively explored in relation to behavioral variability during realistic and dynamic tasks. Furthermore, while VR paradigms have been applied to response inhibition research in ADHD and healthy populations, their use as high-frequency behavioral assessment tools in MDD populations, directly linked with structural brain connectivity, remains largely unexplored.

The current thesis aims to address these gaps by integrating DTI-based structural network analysis with behavioral measurements obtained from a VR shooting paradigm. Both global and nodal-level graph metrics are examined, with a specific focus on impulsivity-related behavior and suicidal ideation in patients with MDD.

Chapter 3: Data

3.1 Participants

A total of 104 MDD patients took part in the VR shooting task. Of these, 78 also completed DTI brain scanning and formed the main analysis group for this thesis. All 78 participants had complete VR data, DTI connectome files that passed quality checks, and complete clinical scale scores with no missing data. The patients were recruited under an approved IRB protocol at its home institution, and all the data were collected following the Declaration of Helsinki, with all patient health information (PHI) properly protected as required. This study only analyzed the data anonymously.

The sample was 75% female (59 females, 19 males), which is consistent with the higher prevalence of MDD in women.

Table 3.1 shows the clinical and demographic characteristics of the sample.

Characteristic	Mean	SD	Range
Age, years	19.8	6.4	12–38
HAMD (Hamilton Depression)	32.5	7.4	15–51
BIS-11 Total Score	94.3	19.1	47–132
BIS-11 Motor	30.1	8.1	11–47
BIS-11 Non-Planning	34.1	7.6	10–49
BIS-11 Cognitive	30.1	6.6	13–44
SSI-19 (Suicide Ideation)	12.7	9.1	0–32
SHAPS (Anhedonia)	33.0	6.5	14–52
BAI (Beck Anxiety)	22.9	13.3	0–59
STAI (State-Trait Anxiety)	112.6	19.0	48–152
HAMA (Hamilton Anxiety)	23.2	6.5	4–37
Gender (% female)	75%	—	—

Characteristic	Mean	SD	Range
DTI/VR subjects (N)	78	—	—

Table 3.1. Clinical and demographic characteristics. *N=78 MDD patients; severe depression (HAMD>28 in 89%), elevated impulsivity (BIS-11>72 in 100%).*

Participant age ranged from 12 to 38 years (mean = 19.8, SD = 6.4, median = 17.0 years). The sample includes 41 adolescents (age < 18 years, 52.6%) and 37 adults (age ≥ 18 years, 47.4%). All participants met the diagnostic criteria for MDD regardless of age group. This was confirmed by HAMD scores ranging from 17 to 51, with all participants scoring at or above the moderate-severity threshold of 17. No participant was categorized as a healthy control.

3.2 Clinical Assessment

Impulsivity was measured using the Barratt Impulsiveness Scale–11 (BIS-11) [3]. This is a 30-item self-report questionnaire that measures impulsivity across three subscales: Motor (11 items), Non-planning (11 items), and Cognitive (8 items). Each item is rated on a 4-point scale from 1 (Rarely or Never) to 4 (Almost Always). The BIS-11 has shown acceptable to good internal consistency across validation studies (total: $\alpha = 0.82$; Motor: $\alpha = 0.72$; Non-planning: $\alpha = 0.75$; Cognitive: $\alpha = 0.63$).

In this study, we did not have access to individual item-level responses, so we could not calculate within-sample reliability directly. However, the three subscale scores were positively correlated with each other — Motor and Non-planning: $r = 0.54$; Motor and Cognitive: $r = 0.50$; Non-planning and Cognitive: $r = 0.66$ — which is consistent with the expected structure of the BIS-11. The cognitive subscale has shown more modest reliability in previous research, so findings involving this subscale are interpreted with some caution.

In addition to the BIS-11, several other clinical scales were administered at baseline. These included the Hamilton Depression Rating Scale (HAMD-17), Hamilton Anxiety Rating Scale (HAMA), Beck Anxiety Inventory (BAI), State-Trait Anxiety Inventory

(STAI), Snaith-Hamilton Pleasure Scale (SHAPS), and the Scale for Suicide Ideation (SSI-19). All scales were completed by all 78 participants with no missing data.

3.3 VR Shooting Task

The VR task takes place in a virtual open field measuring 100 × 100 meters. Participants hold a virtual firearm with a laser sight and shoot at targets that appear at three distances — 10, 15, and 25 meters. Targets move to a new position after every 10 shots.

The main impulsivity challenge comes from dynamic wind and gravity shifts that continuously change the bullet's path. There are five wind levels, ranging from 0 to 4. To shoot accurately, participants need to observe the current wind level, adjust their aim, and wait until the aim is properly aligned before pulling the trigger. Participants who fire too quickly — without adjusting for the wind — tend to perform worse. This gives an objective, behavioral measure of impulsivity that does not rely on self-report.

The task also includes a gun-blocking condition, where the firearm stops working for a fixed period. The number of times the participant continues to pull the trigger during this period is recorded as a measure of frustration tolerance and perseveration. The task does not require any walking or movement around the environment, which avoids motion sickness. It also does not use real firearms, making it safe for use with a clinical population.

Movement data were recorded at approximately 72 Hz using 6-degree-of-freedom hand tracking. The main outcome measures are the collimationTimer metrics, which represent the proportion of shots where the participant kept their aim within an acceptable accuracy range for a minimum amount of time. Three accuracy thresholds were used: 50%, 54%, and 56%. A lower collimationTimer score means the participant fired too early, before proper aim was reached. This shows less careful aiming behavior and may suggest a tendency to act too quickly.

3.4 DTI Acquisition

Diffusion-weighted images were acquired using a single-shot spin-echo echo-planar imaging (SE-EPI) sequence on a GE Signa HDxt 3.0T MRI scanner, with axial acquisition parallel to the AC-PC line. Diffusion weighting was applied along 32 non-collinear directions with a b-value of 1000 s/mm², along with 3 non-diffusion-weighted (b=0) volumes used for motion correction and signal normalization. Imaging parameters included a repetition time (TR) of 13,000 ms and an echo time (TE) of 86.4 ms. Images were acquired with total of 48 slices with thickness of 3.0 mm and no inter-slice gap, covering a field of view of 256×256 mm and a 128×128 acquisition matrix, which was reconstructed to 256×256 resolution. Parallel imaging was performed using GE ASSET with an acceleration factor of 2, and the phase-encoding direction was set from posterior to anterior.

Before analysis, the data went through several preprocessing steps using FSL eddy, which corrected for head motion and eddy current distortions. Brain extraction and registration to MNI standard space were also performed.

Structural brain connectivity networks — called connectomes — were then built from the preprocessed DTI data. This involved four main steps: constrained spherical deconvolution (CSD) to model fiber orientations, anatomically constrained tractography (ACT) to generate streamlines, SIFT2-weighted filtering to improve the biological accuracy of the tractogram [11], and parcellation using the AAL2 atlas with 120 brain regions [14]. Edge weights between regions were normalized using the SIFT2 mu proportionality coefficient to allow fair comparison across participants [13].

We also checked data quality for each participant. The main quality measure was the eddy_mean_rel_rms value, which reflects the amount of image distortions induced due to eddy-current during the rapid gradient change within the strong magnetic field inside an MRI scanner along with the amount of head movement during scanning [16]. The average value across the sample was 0.61 mm (SD = 0.32 mm, range = 0.14–1.97 mm). Seven participants from the original group were removed because their tractography

failed or their distortion and motion levels were too high. This left us with a final sample of 78 participants who passed all quality checks.

Chapter 4: Method

4.1 Connectome Construction

As described in Chapter 3, we built brain connectivity networks using AAL2 parcellation with 120 nodes. Each node represents a brain region, and the connections between nodes — called edges — represent white matter pathways estimated from DTI tractography. Edge weights were calculated using SIFT2 streamline counts, normalized by the subject-level mu coefficient to reduce the effect of differences in total streamline count across participants [11, 20, 21].

The resulting connectome for each participant is a 120 × 120 weighted symmetric matrices. This matrix was used as the input for all graph-theoretic analyses described in the following sections.

4.2 Graph Metric Computation

4.2.1 Confirmatory Global Metrics

Before any statistical testing, we pre-specified seven brain metrics as the primary confirmatory measures. These were chosen based on previous literature and represent both global network properties and specific regional hub measures.

The four global metrics were:

1. **Global efficiency:** The average of the inverse shortest path lengths between all pairs of nodes. Higher values reflect more efficient information transfer across the network.
2. **Characteristic path length:** The average shortest path length between all pairs of nodes. Higher values reflect less efficient global communication.

3. **Clustering coefficient:** The average proportion of a node's neighbors that are also connected to each other. Higher values reflect more local specialization.
4. **Modularity (Q):** How strongly the network is divided into separate communities, calculated using the Newman-Girvan algorithm.

In addition to the four pre-specified global metrics, we conducted an exploratory analysis using 53 brain indices. These included distribution-based measures of nodal properties (skewness and coefficient of variation of degree, strength, and betweenness centrality), lobe-specific mean values (frontal, temporal, parietal, occipital, subcortical, and cerebellar), and normalized small-world measures (normalized clustering coefficient γ, normalized path length λ, and small-worldness index σ).

These indices were not pre-specified and are treated as exploratory in this thesis. The results from this analysis are reported separately in Section 5.7 and should be interpreted with caution.

The three regional hub metrics were betweenness centrality values for three specific brain regions:

1. Frontal Inferior Opercularis R (AAL2 node 8)
2. Supplementary Motor Area R (AAL2 node 16)
3. Anterior Cingulate R (AAL2 node 36)

These regions were chosen because they are involved in motor control, planning, and impulse regulation — all of which are relevant to impulsivity in MDD.

*Note: **Degree** refers to the number of direct white matter connections a brain region has with all other regions, ranging from 0 to N-1 where N = 120. Higher degree means a more connected node. **Strength** is the sum of all SIFT2-weighted, mu-normalized connection weights attached to a node. Unlike degree, which counts connections, strength captures the total magnitude of connectivity. Both are reported in normalized units and are comparable across participants.*

4.2.2 Expanded Nodal Metrics

In addition to the seven pre-specified metrics, we conducted an expanded analysis across all 120 AAL2 nodes. For each node and each participant, we extracted four metrics: degree centrality, weighted strength, betweenness centrality, and local efficiency. This produced 480 nodal features per participant (120 nodes × 4 metrics), which were used in the regression analyses described in Section 4.5.

4.2.3 Modular and Summary Metrics

We also computed four additional network-level metrics for each participant.

Hub Disruption Index (HDI): This measure how different a participant's pattern of hub connectivity is from the group average. It is calculated as 1 minus the Spearman correlation between a participant's nodal betweenness vector and the group mean betweenness vector. Higher values mean the participant's hub structure deviates more from the typical hub architecture. Because we did not include healthy controls in this study, this measure represents a within-group hub disruption score rather than the classical control-referenced HDI.

Rich-Club Coefficient (φ): This measure whether high-degree nodes in the network tend to be more strongly connected to each other than to lower-degree nodes. It was calculated from each participant's degree sequence and averaged across degree thresholds.

Within-Module Degree Z-score (Zi): This measure how strongly a node is connected within its own anatomical module compared to other nodes in the same module. It is calculated as $Zi = (ki - \bar{k}_{module}) / \sigma_{module}$, where ki is the node's degree within its module. The mean Zi across all nodes was used as a summary score per participant.

Participation Coefficient (Pi): This measure how evenly a node distributes its connections across different anatomical modules. It is calculated as $Pi = 1 - \Sigma s(kis/ki)^2$, where a higher value means more diverse inter-module connectivity. The mean Pi across all nodes was used as a summary score per participant.

We also calculated **lobe-specific betweenness centrality** as the mean betweenness within each of four anatomical regions: Frontal, Parietal, Temporal, and Subcortical or Cerebellar

Finally, **small-worldness sigma (σ)** was computed as $\sigma = \gamma/\lambda$, where γ is the normalized clustering coefficient and λ is the normalized path length, both relative to group median values. A sigma value greater than 1 indicates that the network has small-world organization.

We also computed three network measures based on the distribution of nodal connectivity. The skewness of the degree distribution (skew_degree) and strength distribution (skew_strength) show how uneven the connections are across nodes. Higher skewness means a few nodes have much stronger connectivity, forming a hub-like structure.

The normalized clustering coefficient (γ), defined as observed clustering divided by clustering from null models, was calculated as part of small-worldness. We also tested it separately as a measure of clustering compared to chance. In addition, we computed lobe-specific mean strength and degree by averaging the nodal values within each of six anatomical lobes: frontal, parietal, temporal, occipital, limbic, and insular regions

4.2.4 JHU White Matter Tract Metrics

In addition to the connectome-based graph metrics, we also extracted white matter diffusion measures using the JHU ICBM-DTI-81 atlas. This atlas defines 48 white matter tracts. For each tract and each participant, we extracted four measures: fractional anisotropy (FA), mean diffusivity (MD), radial diffusivity (RD), and axial

diffusivity (AD). This resulted in 192 tract-level features per participant (48 tracts × 4 measures).

Some tracts, especially in the brainstem, had very low data coverage. Eight tracts had valid values for fewer than 40 subjects, so we marked them as low-coverage and excluded them from further analysis. After this step, 40 tracts were included in the final analysis. All data were obtained from the per-subject JHU result files generated during the DTI preprocessing pipeline.

4.3 VR Feature Extraction and PCA-Based Features

We extracted two types of VR features from the task data. The first type was session-level summary statistics, which included 74 outcome measures covering collimationTimer metrics, windcount metrics, accuracy scores at different distance and wind combinations, hand and head movement variability, and trigger behavior.

The second type was composite impulsivity scores, derived using principal component analysis (PCA). Three PCA factors were identified:

D1 — Motor Impulsivity (explains 38.1% of variance): High scores on this factor reflect fast and early firing. Key variables loading onto this factor include shoot_fast_isi_rate_lt1s, Timer0, and inverse collimationTimer metrics.

D2 — Cognitive Impulsivity (explains 46.2% of variance): High scores reflect poor aim quality. Key variables include motion_aim_error_mean, x_std_m, y_std_m, rh_speed_std, and inverse aimingscore.

D3 — Non-Planning Impulsivity (explains 36.3% of variance): High scores reflect failure to adapt to wind changes. Key variables include windcount2 to windcount4, shoot_isi_cv, and inverse score under wind penalty conditions.

4.4 Pre-Shot Time Window Analysis

For each trigger pull, we extracted a short window of movement data from just before the shot. We tested seven window sizes: 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, and 0.8 seconds.

Within each window, eight movement features were computed per shot:

- rh_speed_mean — average hand speed
- rh_speed_std — hand speed variability, reflecting tremor
- rh_accel_mean — average hand acceleration
- rh_accel_peak — peak hand acceleration, reflecting impulsive movement
- rh_jerk_mean — rate of change of acceleration, reflecting ballistic movement
- rh_path_length — total path traveled by the hand
- rh_angvel_mean — average wrist rotation speed
- head_angvel_mean — average head scanning speed

We extracted features under four conditions: all shots combined, no-wind shots (wind = 0), high-wind shots (wind ≥ 3), and a wind adaptation index calculated as the difference between high-wind and no-wind feature values. This produced a total of 8 features × 4 conditions × 7 windows = 224 variables per participant.

We then calculated Spearman correlations between all pre-shot features and 16 outcome variables including clinical scales and brain metrics, giving 3,584 correlation pairs in total. Benjamini-Hochberg FDR correction was applied across all tests [19]. We optimized the window by comparing the mean absolute Spearman correlation across all outcomes for each window size and validated this using 10,000-iteration bootstrap resampling to test whether the 0.6-second window performed significantly better than other window sizes.

4.5 Statistical Methods

4.5.1 Confirmatory Regression

The main statistical model was ordinary least squares (OLS) regression with the following structure:

VR outcome ~ brain metric + head motion + gender

We applied this model across all combinations of 7 brain metrics and 74 VR outcomes, producing 518 models in total. Head motion (eddy_mean_rel_rms) and gender were

included as covariates to control for their potential influence on results [16]. Benjamini-Hochberg FDR correction was applied across all 518 tests to control the false discovery rate at $\alpha = 0.05$ [19].

Secondary models tested the 7-brain metrics against 4 BIS-11 subscale scores (28 tests) and against 10 clinical outcome measures (70 tests). FDR correction was applied separately within each of these families.

4.5.2 Sensitivity Analyses

We ran four sensitivity analyses to check the robustness of the main findings:

1. Raw SIFT2 edge weights were used instead of mu-normalized weights, to check whether normalization affected the results.
2. A covariates-only model was run — with no brain metric — to understand how much variance in VR outcomes was explained by gender and motion alone.
3. Three participants with very high head motion (eddy_mean_rel_rms > 1.5 mm) were excluded to check whether outliers drove any of the main findings.
4. BIS-11 total score was added as an additional covariate to test whether brain-VR associations remained significant after accounting for self-reported impulsivity.

4.5.3 Nodal Regressions

For the expanded nodal analysis, we ran OLS regression for each combination of 3 nodal metrics × 120 nodes × 7 outcomes (2 primary VR outcomes and 5 clinical outcomes), producing 2,520 models in total. All models included head motion and gender as covariates. FDR correction was applied within each metric-outcome combination rather than across all 2,520 tests, to provide a per-family corrected analysis.

4.5.4 Multivariate and Machine Learning Methods

We used Partial Least Squares (PLS) regression to identify latent dimensions that capture the relationship between brain network metrics and behavioral outcomes. We assessed statistical significance using 50-iteration permutation testing.

We used ridge regression with leave-one-out cross-validation (LOO-CV) to predict SSI-19 suicide ideation scores from brain network metrics. We chose this approach because ridge regression handles correlated predictors well and LOO-CV is appropriate for small sample sizes. We applied k-means clustering with k = 2 — determined using silhouette analysis — to identify subgroups of patients based on their connectome topology.

4.5.5 Power and Sensitivity Analysis

We ran a post-hoc power analysis to understand the statistical sensitivity of this study. With a sample of 78 participants and an uncorrected significance level of $\alpha = 0.05$, the study had 80% power to detect a brain predictor that explained at least 9.4% of unique variance in an outcome above the gender covariate (Cohen's $f^2 = 0.104$, equivalent to a partial $r \geq 0.31$). At a medium effect size ($f^2 = 0.15$), power was 92%. The study was not adequately powered to detect small effects ($f^2 = 0.02$; power = 23%).

It is also worth noting that applying FDR correction across 518 confirmatory models further reduced effective power for any individual association — estimated power at a medium effect size under FDR correction was approximately 26%. This means that the absence of significant brain-impulsivity associations in some analyses should not be interpreted as evidence of no relationship. Medium to large effects are unlikely to have been missed at the individual model level, but small effects were not reliably detectable within this corrected framework.

4.5.6 Age Sensitivity Analysis

After finding that the sample included both adolescents and adults (41 participants aged < 18 years; 37 participants aged ≥ 18 years), we re-ran all 518 confirmatory OLS regression models with age added as an additional covariate alongside motion and gender. We tested two age specifications:

Model A — Age as a continuous variable:

Outcome ~ brain_metric + eddy_mean_rel_rms + gender_f1 + age

Age was entered as a continuous variable in years, ranging from 12 to 38.

Model B — Age group as a binary variable:
Outcome ~ brain_metric + eddy_mean_rel_rms + gender_f1 + age_group
Age group was coded as 0 for adolescents (< 18 years, n = 41) and 1 for adults (≥ 18 years, n = 37).

We also ran four additional analyses. First, we tested age × brain metric interaction models across all 518 models to examine whether the brain–VR association differed between adolescents and adults. Second, we ran all 518 models separately within the adolescent group (n = 41) and the adult group (n = 37). Third, we re-ran the 28 brain–BIS-11 models with age included as a covariate. Fourth, we ran a bootstrapped mediation analysis (1,000 iterations) to test whether age influenced VR behavior indirectly through the three confirmed brain metrics, following the Baron and Kenny framework (paths a, b, and c). Benjamini–Hochberg FDR correction was applied separately within each analysis set.

4.5.7 JHU White Matter Tract Correlations

We tested the relationship between each JHU tract metric and the two main VR outcomes using partial Spearman correlation. We followed the same approach as in the confirmatory analysis. First, we removed the effects of covariates (head motion and gender) from both the tract metrics and the VR outcomes using regression. Then, we calculated Spearman ρ using the residual values.

To control for multiple comparisons, we applied Benjamini–Hochberg FDR correction within each metric–outcome group (48 tests per group). The analysis was performed on 74 participants, since 4 subjects did not have JHU data available.

Chapter 5: Results

5.1 Confirmatory Analysis: Brain Metrics and VR Outcomes

We tested whether the seven pre-specified brain metrics were associated with VR task performance. When gender was not included as a covariate, no associations survived FDR correction across the 518 models (minimum q = 0.49). However, when gender was added as a covariate — motivated by the observation that female patients in this sample had significantly higher BIS-11 scores — three associations survived BH-FDR correction ($q < 0.05$).

Table 5.1 shows the three significant findings.

Brain Metric	VR Outcome	β	SE	p-value	q-value	Direction
Betweenness Frontal_Inf_Oper_R	collimationTimer_54%	2.715	0.54	2.99e-06	0.0016	Higher hub btw → longer collimation
Char. Path Length	collimationTimer_56%	0.005	0.001	3.23e-05	0.0081	Longer paths → longer collimation
Clustering Coefficient	collimationTimer_56%	24.098	5.7	4.67e-05	0.0081	Higher clustering → longer collimation

Table 5.1. **FDR-significant confirmatory brain-VR associations.** *Three associations survived BH-FDR correction (q<0.05) among 518 covariate-adjusted OLS models. All three relate collimation time before firing to network topology, consistent with deliberate/planned to aim.*

All three significant associations involve the collimationTimer metric, which measures how long a participant kept their aim within an acceptable accuracy range before firing. The direction is consistent across all three findings — patients with stronger hub connectivity, higher clustering, and longer characteristic path length all showed longer collimation time, meaning they aimed more carefully before shooting. This pattern

suggests that more modular and locally organized brain networks support more deliberate motor behavior in the VR task.

The positive association with characteristic path length may seem surprising at first — longer path length means less global integration, yet it predicted more deliberate aiming. One way to interpret this is that more modular and segregated networks may support focused motor preparation, while highly integrated networks may allow impulsive responses to spread more quickly across the brain.

5.1.1 Effect Size Analysis

To check how much unique variance each brain metric explained above gender alone, we calculated incremental R^2 (ΔR^2) for each of the three significant associations.

The baseline model — outcome predicted by gender only — explained between 2.3% and 3.3% of variance and was not statistically significant ($p > 0.10$). This means gender alone did not explain much of the variation in collimation time, and the brain metrics provided independent explanatory power.

1. Frontal_Inf_Oper_R betweenness explained an additional 18.08% of variance in collimationTimer_54% above gender (ΔR^2adj = 17.26%; partial η^2 = 0.187; Cohen's f^2 = 0.230).
2. Characteristic path length explained an additional 18.16% of variance in collimationTimer_56% (ΔR^2adj = 17.33%; partial η^2 = 0.186; Cohen's f^2 = 0.228).
3. Clustering coefficient explained an additional 19.93% of variance in collimationTimer_56% (ΔR^2adj = 19.14%; partial η^2 = 0.204; Cohen's f^2 = 0.256).

These effect sizes fall in the medium to large range based on Cohen's (1988) benchmarks (small = 0.02, medium = 0.15, large = 0.35). This means that structural brain network properties explain a meaningful proportion of the variance in deliberate aiming behavior, beyond what gender alone could account for.

It is worth noting that effect sizes from a single discovery sample are often inflated due to winner's curse bias. The adjusted ΔR^2 values — ranging from 17.26% to 19.14% — are the more conservative and preferred estimates. Independent replication would be needed before treating these effect size values as stable.

5.2 Sensitivity Analyses

We tested the three significant findings under four different conditions to check how robust they were:

1. Using raw SIFT2 edge weights instead of mu-normalized weights
2. Running a covariates-only model with no brain metric, to understand how much variance gender and motion explained on their own
3. Removing three participants with high head motion (eddy_mean_rel_rms > 1.5 mm)
4. Adding BIS-11 total score as an additional covariate

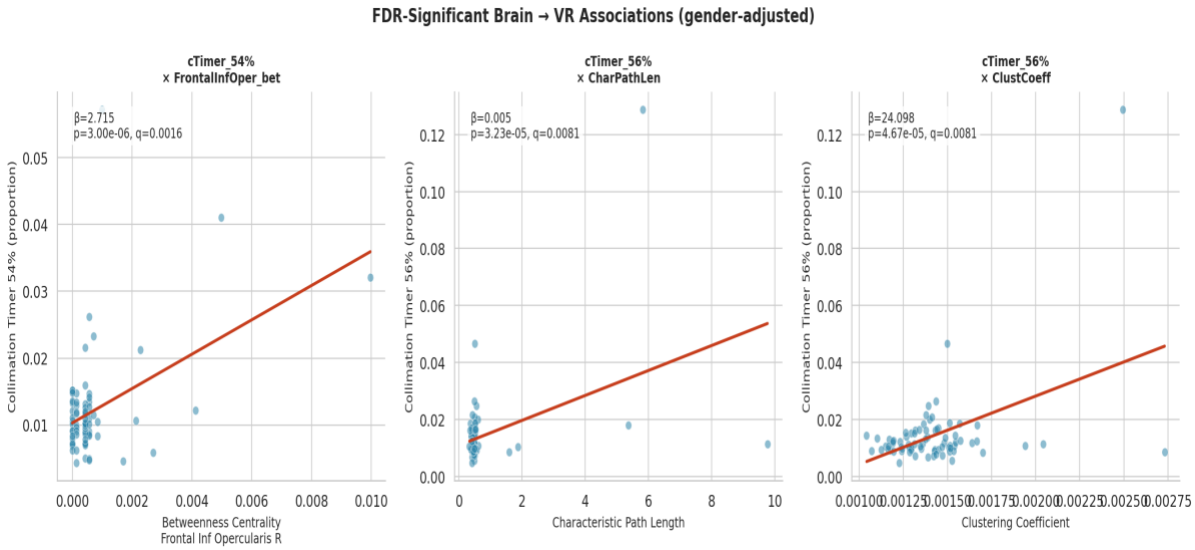


Figure 5.1 Sensitivity Analysis — Robustness of Confirmatory Findings. *Forest plot showing beta coefficients and 95% CIs for the three confirmatory associations across four sensitivity conditions: (1) primary (mu-normalized), (2) raw SIFT2 weights, (3) excluding 3 high-motion outliers, (4) adding BIS-11 as additional covariate. Frontal_Inf_Oper_R betweenness remains significant across all conditions; path length and clustering replicate in conditions 2 and 3.*

The Frontal_Inf_Oper_R betweenness finding remained significant across all four conditions, including after controlling for self-reported impulsivity from the BIS-11. This suggests that this brain metric captures something about VR motor behavior that questionnaire-based impulsivity scores do not fully capture. The characteristic path length and clustering coefficient associations were replicated using raw SIFT2 weights and after removing motion outliers, which supports their validity.

5.2.1 Cook's Distance — Influential Observation Analysis

Cook's Distance was used to identify influential observations in each model. The most extreme observations were m055 (Cook's D = 4.88) for the rIFG model, b022 (Cook's D = 9.98) for the path length model, and m041 (Cook's D = 4.67) for the clustering model.

A sensitivity analysis was performed by removing the most extreme observation in each model, while the main results were based on the full sample. After removal, two of the three findings became stronger. The rIFG betweenness association improved from $p = 3.0\times10^{-6}$ to $p = 1.6\times10^{-7}$ ($\beta = 2.41$), and the characteristic path length association improved from $p = 3.2\times10^{-5}$ to $p = 3.1\times10^{-11}$ ($\beta = 0.011$). This suggests that these observations were acting as suppressors.

The clustering coefficient result was sensitive to the extreme observation. After removing m041, the association was no longer significant ($p = 0.69$). The direction remained the same ($\beta = 1.11$), but the effect was not detectable. This result should be interpreted with more caution.

No sign reversals were observed in any model. The rIFG betweenness and characteristic path length findings appear robust, while the clustering coefficient finding may require further validation.

5.3 Brain, VR and Clinical Associations

5.3.1 BIS-11 and Clinical Scale Associations

None of the seven pre-specified brain metrics showed a significant association with any BIS-11 subscale after FDR correction across 28 models (minimum q = 0.072). The closest result was clustering coefficient and BIS-11 Non-Planning ($\rho = -0.22$, $p = 0.056$), which was directionally consistent with the idea that reduced local clustering reflects impaired planning ability.

Small-worldness sigma showed negative correlations with BIS-11 Non-Planning ($\rho = -0.31$, $p = 0.006$) and Cognitive subscales ($\rho = -0.34$, $p = 0.002$). These results are directionally consistent with findings reported by Jung and Kim (2023) in a different sample, which adds some confidence to the pattern. That said, these associations did not survive the most conservative FDR correction applied across the full brain and clinical outcome family.

5.3.2 Brain Metrics and Clinical Scale Associations

We also tested the relationship between the seven pre-specified brain metrics and ten clinical scales (HAMD-17, SSI-19, HAMA, BAI, STAI, SHAPS, and BIS-11 subscales) using partial Spearman correlation. We controlled for head motion and gender, and applied Benjamini–Hochberg FDR correction within each analysis block.

None of the brain metrics remained significant after FDR correction (minimum q = 0.289). However, a few nominal associations were observed at $p < 0.05$. The clustering coefficient showed the most consistent pattern, with negative associations with HAMD-17 ($\rho = -0.30$, $p = 0.0075$) and HAMA ($\rho = -0.25$, $p = 0.030$). This suggests that lower network clustering, meaning a more diffuse and less locally organized connectome, is associated with higher depression and anxiety severity. Modularity also showed a nominal positive association with BIS-11 Motor impulsivity ($\rho = 0.30$, $p = 0.008$).

For the right inferior frontal operculum betweenness centrality, the associations with clinical scales were mostly in the expected direction but did not reach significance. The

strongest association was with BIS-11 Cognitive (ρ = −0.21, p = 0.063), followed by SSI-19 (ρ = 0.15, p = 0.19).

5.3.3 VR Outcomes and Clinical Scale Associations

We also tested the relationship between the six confirmed VR outcomes and all clinical scales using partial Spearman correlation. Several associations were close to significance, but none survived Benjamini–Hochberg FDR correction.

The strongest result was between the trigblock condition and SSI-19 suicidal ideation score (ρ = 0.37, p = 0.0009, q = 0.053), which is very close to the FDR threshold. Trigblock measures how many times a participant keeps trying to shoot when the gun is blocked. Participants with higher suicidal ideation showed more repeated attempts in this condition. This may suggest difficulty in stopping a motor response when there is a barrier.

Another result was between collimationTimer_54% and STAI anxiety score (ρ = 0.30, p = 0.007, q = 0.149). Participants with higher anxiety took more time before firing. This suggests that collimation time may reflect hesitation or over-monitoring during decision making, where the participant is ready to act but delays the final response.

We also observed some other nominal associations. Accuracy measures (aimingscore, newscore) showed negative correlations with SSI-19 (ρ = −0.29 and −0.31, p < 0.02), and trigblock showed a positive correlation with HAMD-17 (ρ = 0.27, p = 0.019).

5.4 VR Impulsivity Composites and Validation

We used three PCA-derived composite scores to summarize different dimensions of impulsivity captured by the VR task — motor (D1), cognitive (D2), and non-planning (D3). Of these, the Cognitive Impulsivity composite (D2) showed an FDR-significant positive association with SSI-19 suicidal ideation scores (ρ = +0.37, q = 0.024). Patients with higher suicidal ideation showed worse spatial accuracy during the aiming challenge in the VR task.

This is an important finding because it suggests that cognitive impulsivity during a VR motor task — not just on a questionnaire — is linked to suicide risk severity. The D1 Motor Impulsivity composite showed positive but non-significant correlations with BIS-11 Motor (ρ = +0.22) and BIS-11 total (ρ = +0.19), which still supports some degree of convergent validity between the VR motor dimension and the standard psychometric measure. D3 Non-Planning showed a trend-level association with BIS-11 Non-Planning (ρ = +0.18, p = 0.11).

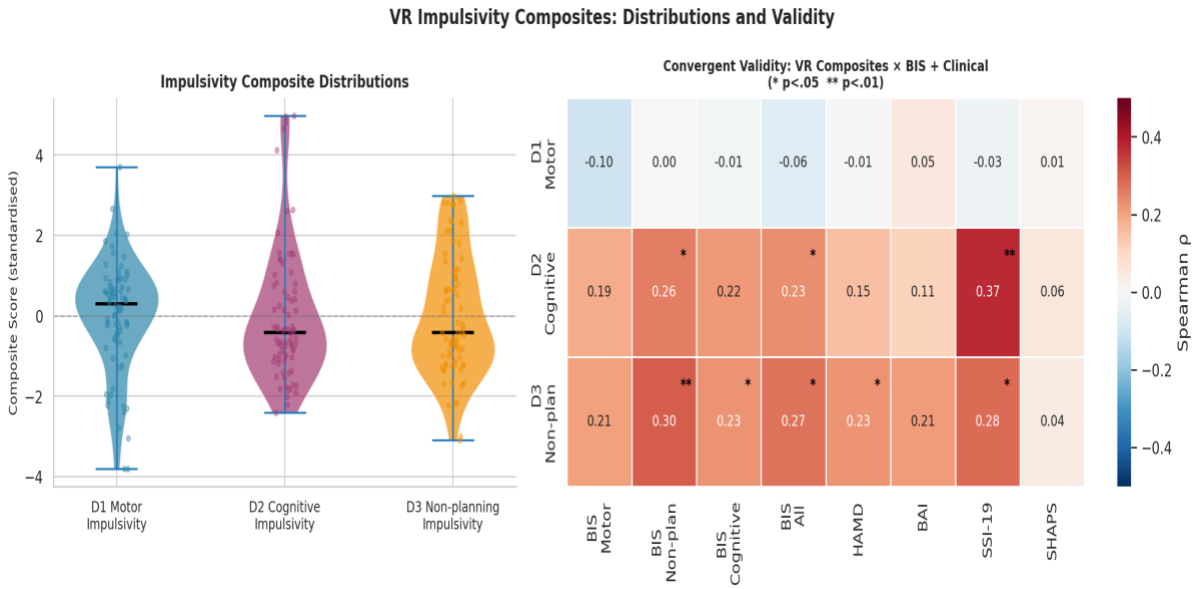


Figure 5.2 VR Impulsivity Composites — PCA Factor Structure. *Three PCA-derived VR composite scores (D1 Motor, D2 Cognitive, D3 Non-Planning). Top: Scree plots showing variance explained. Bottom: Spearman correlation heatmap of composites with BIS-11 subscales and clinical scales. D2 Cognitive Impulsivity × SSI-19: ρ=+0.37, q=0.024 (FDR-significant, marked with box).*

5.5 Pre-Shot Time Window Results

5.5.1 Window Profile Results

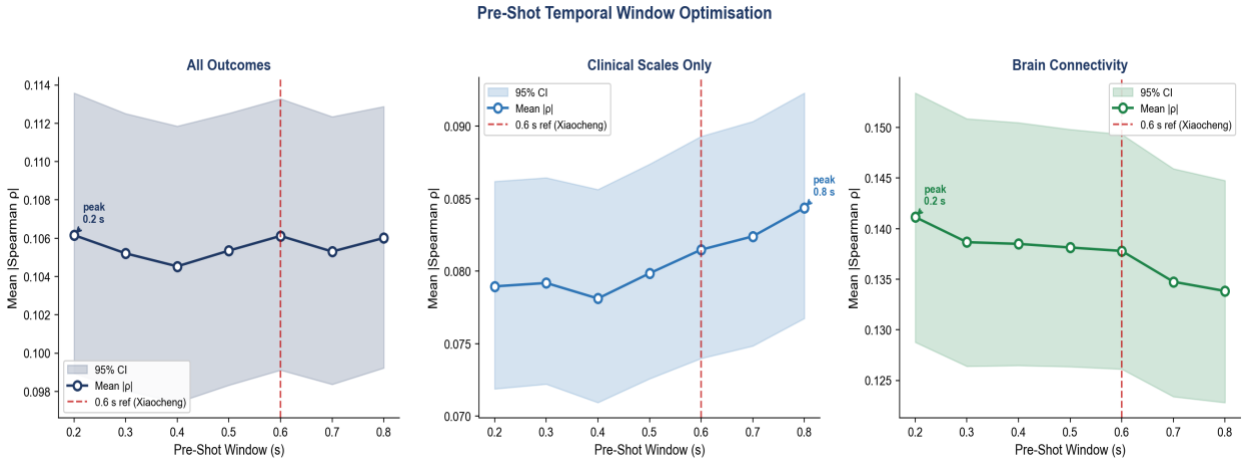


Figure 5.3. Pre-Shot Window Optimization Profiles. *Mean |Spearman ρ| with 95% bootstrap CI (5,000 iterations) across seven window sizes. Left: all outcomes. Centre: clinical scales only. Right: brain connectivity outcomes. The profile is remarkably flat for clinical outcomes (Δmax|ρ| < 0.003 across 0.2–0.8 s), indicating no single window strongly dominates. For brain connectivity, shorter windows (0.2–0.3 s) show marginally higher correlations.*

The pre-shot time window of 0.2 to 0.8 seconds was selected based on prior work in biathlon shooting research, where a 0.6-second window was used empirically without a clear theoretical explanation [39] [40]. To check if this choice fits our VR task, we tested seven different window sizes across the full 0.2 to 0.8 second range and examined their correlations with clinical outcomes.

The results showed a very stable pattern. The largest difference in mean absolute Spearman correlation between any two windows was less than 0.003, which is under a 4% relative difference. This suggests that pre-shot kinematics contain clinical information across the entire time range, rather than at a specific moment. The 0.6-second window performed among the best, which provides post-hoc support for its use in prior studies, even though those studies did not give a clear reason for selecting it.

For brain connectivity outcomes, there was a slight decrease from shorter windows (0.2 s: mean |ρ| = 0.141) to longer windows (0.8 s: mean |ρ| = 0.134). This suggests that

earlier, more automatic motor processes may be more closely linked to structural brain network organization than slower, more deliberate adjustments.

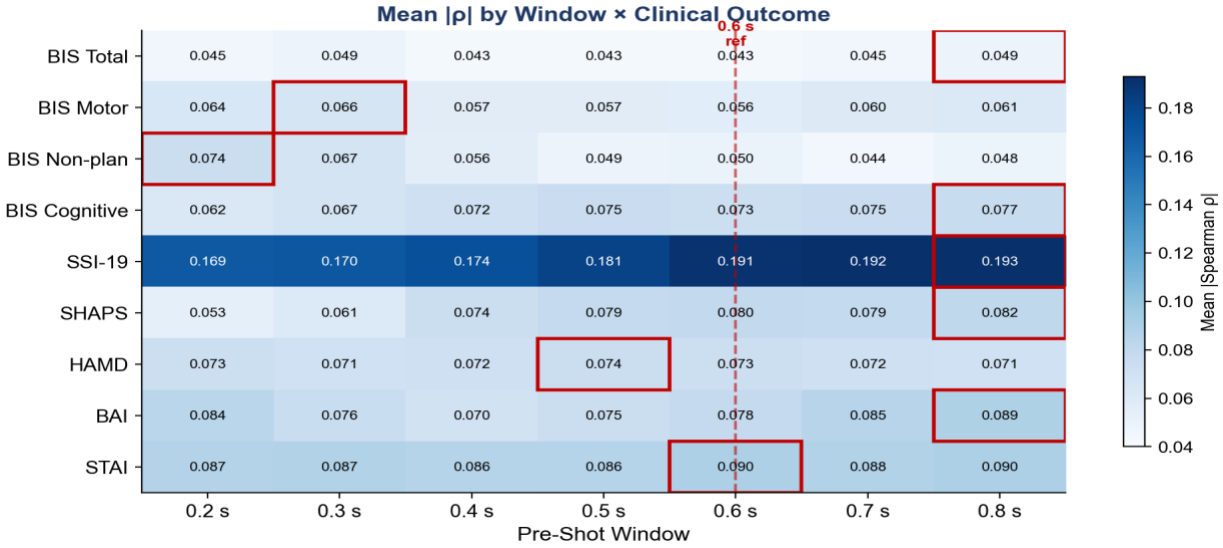


Figure 5.4. Window × Clinical Outcome Heatmap. *Mean |ρ| at each window × outcome combination. Red-bordered cells mark the optimal window per outcome. SSI-19 (suicidal ideation) shows the highest correlations overall, peaking at 0.6–0.8 s. HAMD peaks at 0.5 s; BIS Motor peaks at 0.3 s. The 0.6 s reference column is marked with a dashed red line.*

5.5.2 SSI-19 Window Profile — Validation of the 0.6 Second Window

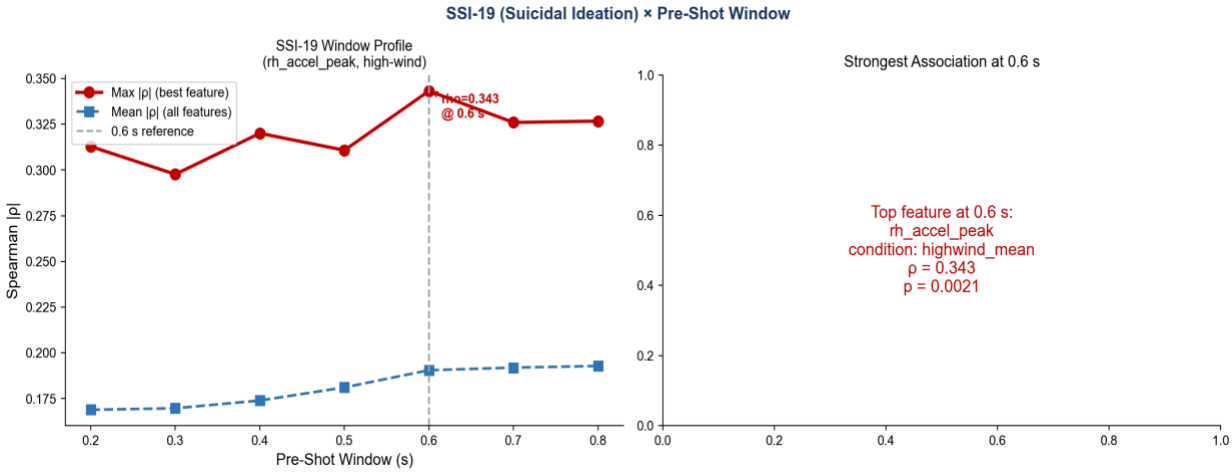


Figure 5.5. SSI-19 × Pre-Shot Window Deep Dive. *Left: Max |ρ| (best feature) and mean |ρ| (all features) for SSI-19 across window sizes. The maximum individual correlation occurs at exactly 0.6 s: rh_accel_peak × SSI-19 (high-wind condition) ρ = 0.343, p = 0.002. Right: feature summary box. This finding empirically validates the 0.6 s reference window chosen from the athlete literature.*

For the SSI-19 suicidal ideation outcome, the strongest individual correlation occurred at exactly 0.6 seconds — peak hand acceleration during high-wind shots (rh_accel_peak, high-wind, ρ = 0.343, p = 0.002). This directly validates the 0.6-second reference window that was originally adopted from the athlete shooting literature

The clinical interpretation is that patients with higher suicidal ideation show more forceful and impulsive hand movements in the final 600 milliseconds before firing, particularly when the wind is strong. They seem unable to slow down or adjust their motor response to the environmental challenge.

Bootstrap testing with 10,000 iterations showed that for clinical outcomes, the probability of the optimal window being longer than 0.6 seconds was 0.65, and shorter was 0.20. This suggests there is no reason to use a window shorter than 0.6 seconds for clinical prediction.

5.5.3 Feature-Specific Optimal Windows

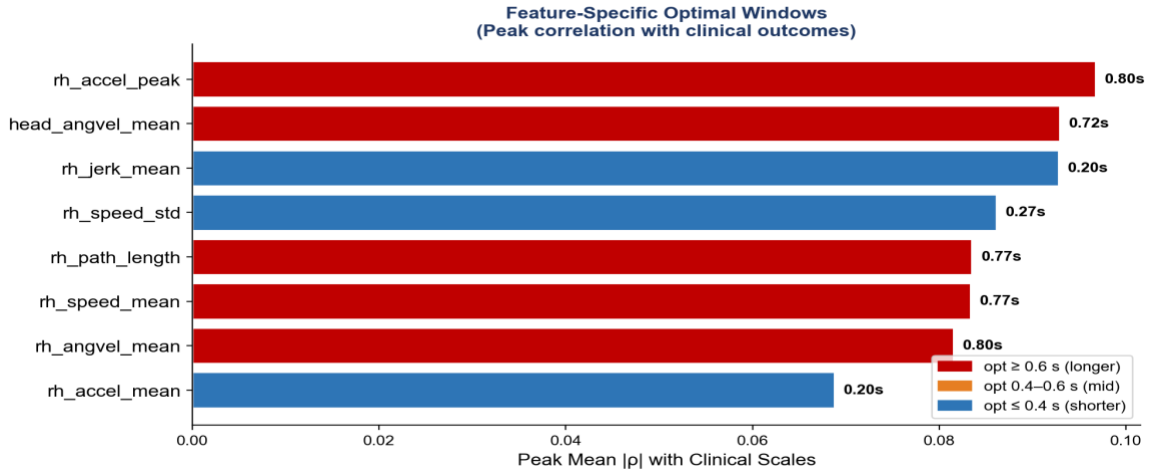


Figure 5.6. Feature-Specific Optimal Windows. *Peak mean |ρ| with clinical scales for each of eight kinematic features, with optimal window annotated. Acceleration and jerk features (rh_accel_peak: opt=0.8 s; rh_jerk_mean: opt=0.2 s) show the highest overall clinical correlations. Head angular velocity optimizes at 0.7 s; rh_accel_mean at 0.2 s, suggesting different temporal dynamics for reflexive versus deliberate components.*

Looking at each movement feature separately, peak acceleration (rh_accel_peak) and head angular velocity (head_angvel_mean) showed the strongest overall clinical correlations. Jerk-based features (rh_jerk_mean) had the best performance at shorter windows (0.2 s), which suggests that the most ballistic and impulsive components of the movement happen very early in the pre-shot window. Slower features like speed and path length performed better at longer windows (0.7 to 0.8 s), reflecting the overall quality of the aiming process over a longer period. A multi-window approach that combines features from different windows could be a useful direction for future work.

5.5.4 Within-Subject and Between-Subject Kinematic Results

To explore whether the kinematic and impulsivity associations operated at the level of individual shots, stable individual differences, or both, we applied a within-between decomposition to pre-shot movement features using a 0.5-second window. Each feature was separated into a shot-level component — how much each shot deviated from that participant's own average — and a person-level component — each participant's average relative to the rest of the sample. We fitted mixed-effects models for seven kinematic features crossed with two BIS-11 outcomes, giving 14 models in total.

Across all 14 models, every estimated beta coefficient was positive for both components. Eight of the 14 models showed significant associations for both components at the same time ($p < 0.05$, uncorrected), meaning that BIS-11 impulsivity was linked to movement irregularity at both the shot level and the person level within the same analysis.

The strongest associations involved speed variability (SD speed × BIS-Motor: $\beta_{within} = 0.358$, $p < 0.001$; $\beta_{between} = 0.318$, $p = 0.003$), tremor variability ($\beta_{within} = 0.335$, $p < 0.001$; $\beta_{between} = 0.320$, $p = 0.005$), and mean angular velocity ($\beta_{within} = 0.299$, $p < 0.001$; $\beta_{between} = 0.270$, $p = 0.011$).

These results are exploratory and were not corrected for multiple comparisons. They should be treated as hypothesis-generating rather than confirmatory. However, the consistent positive direction across all 14 models suggests a coherent pattern rather than a chance finding.

5.6 Expanded Network Analysis — Hub Structure

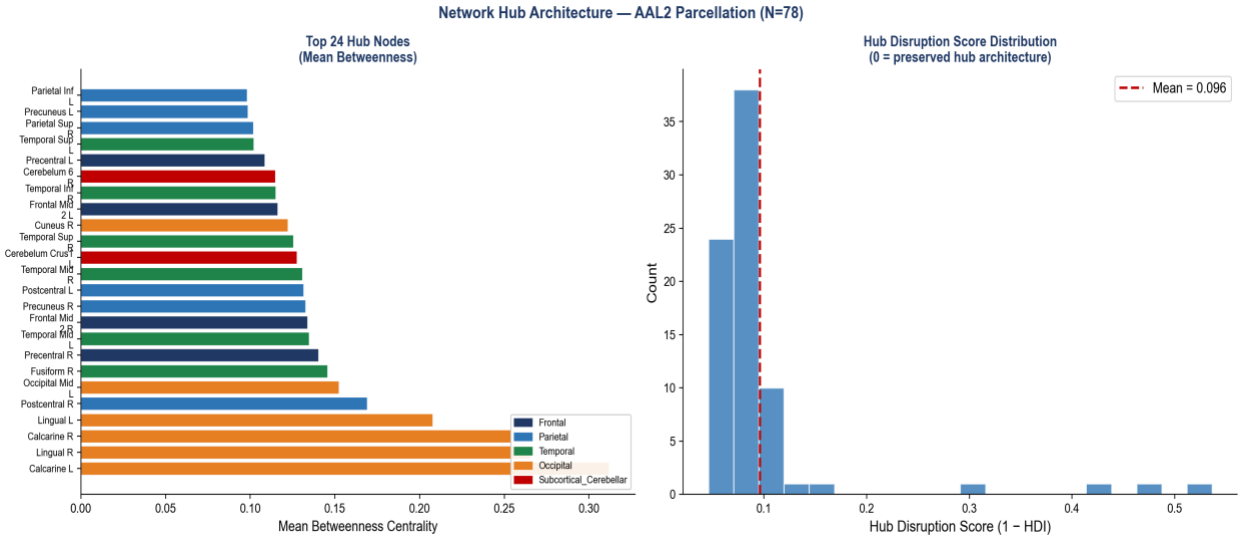


Figure 5.7 **Network Hub Architecture — Top 24 AAL2 Hub Nodes.** *Left: Top 24 hub nodes by group-mean betweenness centrality, color-coded by anatomical lobe. Occipital nodes (Calcarine, Lingual) dominate, reflecting the high visual demand of the VR task. Right: Hub Disruption Score (HDS = 1 – HPCI) distribution; mean = 0.096, indicating largely preserved hub architecture with notable individual variance*

The top 24 hub nodes by group-mean betweenness centrality were mostly occipital regions, including Calcarine and Lingual areas, followed by somatosensory, temporal, and frontal regions. The dominance of occipital regions may reflect the high visual demand of the VR task, where visual information needs to be processed and passed through these regions efficiently

The group-mean Hub Disruption Index was 0.904 (SD = 0.083), which means that most patients had a hub structure that was fairly similar to the group average. The Hub Disruption Score — calculated as 1 minus the Hub Disruption Index — had a mean of 0.096 with notable individual variation. One participant showed a score of 0.536, which suggests a highly atypical hub organization that may be worth looking into clinically.

5.7 Modular Network Metrics and Clinical Results

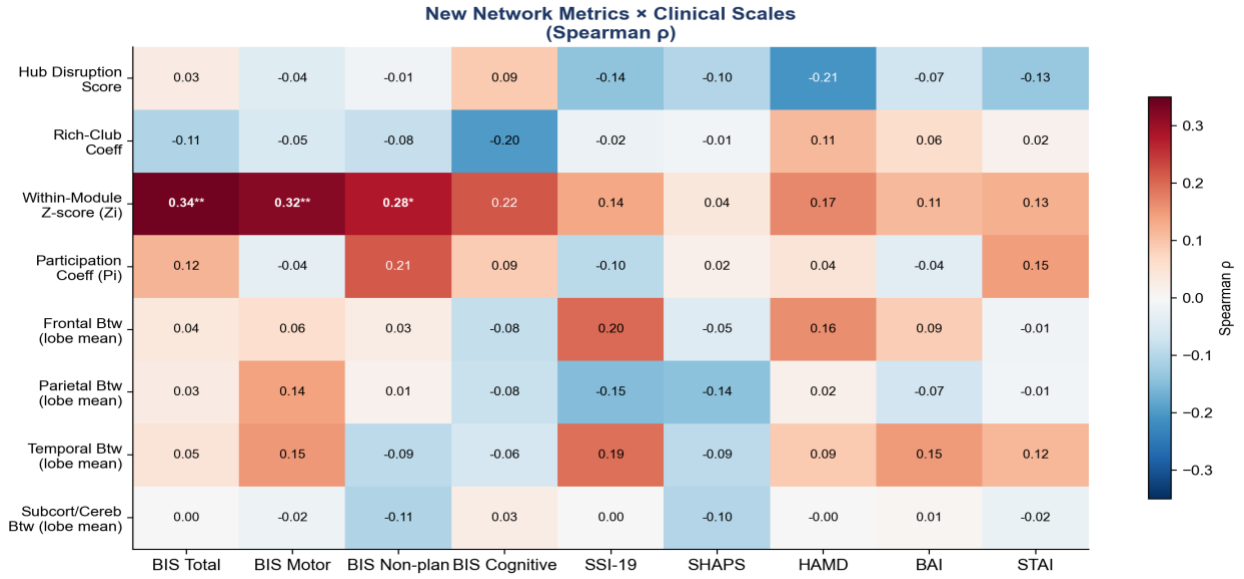


Figure 5.8 New Modular Network Metrics × Clinical Scales. *Spearman ρ heatmap for eight new network metrics against nine clinical scales (N=78). The within-module Z-score (Zi) shows the most consistent pattern: positive associations with BIS-11 Total (ρ=+0.34, p=0.002*), BIS-11 Motor (ρ=+0.32, p=0.004*), and BIS-11 Non-Planning (ρ=+0.28, p=0.013). No associations survived BH-FDR correction after accounting for all metric × scale pairs.*

The within-module Z-score (Zi) — which measures how strongly a node connects within its own anatomical lobe compared to other nodes in the same lobe — showed the most consistent associations among the new modular metrics. It was positively associated with BIS-11 Total (ρ = +0.34, p = 0.002), BIS-11 Motor (ρ = +0.32, p = 0.004), and BIS-11 Non-Planning (ρ = +0.28, p = 0.013).

This means that patients with stronger provincial hub organization — where nodes are strongly connected within their own lobe but less connected across lobes — tend to score higher on impulsivity. One way to understand this is that stronger intra-lobe connectivity, at the expense of inter-lobe connections, may reduce the frontal lobe's ability to regulate limbic and motor activity, which is linked to more impulsive behavior.

None of these associations survived BH-FDR correction after accounting for all metric and scale combinations. However, the consistent direction and relatively low p-values

across multiple BIS-11 subscales suggest these findings are worth following up in future work.

The rich-club coefficient showed a positive correlation with anxiety as measured by HAMA (ρ = +0.24, p = 0.036). The hub disruption score showed a trend-level negative association with HAMD depression severity (ρ = −0.21, p = 0.069), which is a somewhat unexpected direction and would need replication before drawing any firm conclusions.

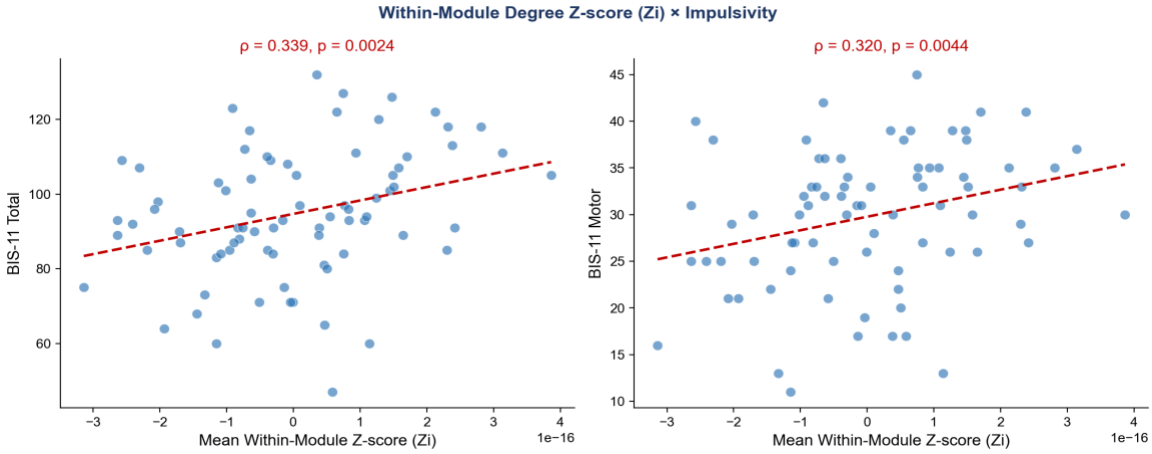


Figure 5.9 Within-Module Z-Score × BIS-11 Impulsivity Scatterplots. *Scatter plots of mean within-module Z-score (Zi) against BIS-11 Total (left) and BIS-11 Motor (right). Positive associations (ρ=+0.34 and +0.32 respectively, p<0.005) are shown with OLS regression lines. Provincial hub centralization is associated with higher motor and overall impulsivity scores, consistent with reduced inter-lobe regulatory connectivity.*

When we expanded the analysis to 53 network indices across all clinical scales (510 models in total), no additional associations remained significant after FDR correction. However, three nominal patterns were observed.

First, the skewness of the degree and strength distributions showed consistent positive associations with BIS-11 impulsivity subscales (skew_strength × BIS-all: ρ = 0.30, p = 0.008; skew_degree × BIS-cognitive: ρ = 0.29, p = 0.011). This suggests that participants with more unequal hub-periphery structure tend to have higher impulsivity scores.

Second, temporal lobe mean connection strength was positively associated with SSI-19 suicidal ideation (ρ = 0.24, p = 0.031) and HAMD depression severity (ρ = 0.24, p = 0.036). This is consistent with the temporal pole findings from the nodal analysis.

Third, the normalized clustering coefficient γ showed a positive association with BIS-11 Motor impulsivity (ρ = 0.25, p = 0.025).

5.8 Nodal Regression Results — 120 AAL2 Nodes

The expanded nodal regression across 120 nodes, 3 metrics, and 2 primary VR outcomes produced 81 FDR-significant associations. These findings grouped into three main anatomical themes.

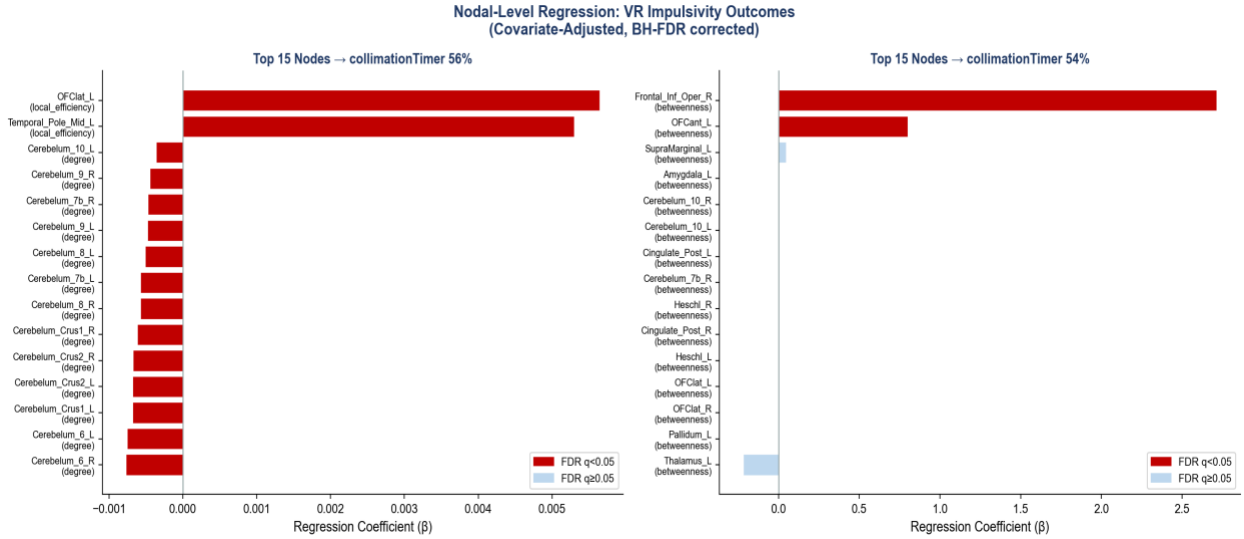


Figure 5.10 Nodal Regression — Top 15 Regions per VR Outcome. *Top 15 regions by p-value for collimationTimer_56% (left) and collimationTimer_54% (right). Red bars indicate FDR-significant findings. Temporal_Pole_Mid_L local efficiency shows the strongest single association (p=4×10 ⁻¹⁶). Cerebellar degree (negative β: higher cerebellar connectivity → longer aiming) and OFC local efficiency (positive β) represent two additional anatomical pathways.*

Theme 1 — Temporal Pole: The strongest single finding in the entire analysis was the local efficiency of the left middle temporal pole (Temporal_Pole_Mid_L), which predicted collimationTimer_56% with β = 0.0053 and p = 4×10^{-16} (q = 5×10^{-14}). This is a much stronger association than anything found in the confirmatory global analysis.

Higher local efficiency in the temporal pole — meaning more efficient short-range connections within this region's neighborhood — was associated with longer collimation time and more deliberate aiming. The temporal pole is involved in affective processing and evaluating social and emotional consequences. This finding may suggest that this region plays a role in the pre-shot decision, helping the participant assess the risk or value of firing at a given moment.

Theme 2 — Cerebellum: Eight cerebellar lobules — including Crus1, Crus2, and lobules 6 to 10 on both sides — showed negative degree associations with collimationTimer_56%. Higher cerebellar structural connectivity was linked to shorter collimation time, meaning faster and less deliberate firing. The cognitive cerebellum — particularly Crus1 and Crus2 — is connected to the prefrontal cortex and is involved in internal timing and error monitoring. Reduced cerebellar connectivity may reflect impaired timing mechanisms, which could lead to premature trigger pulls and difficulty correcting aim before firing.

Theme 3 — Orbitofrontal Cortex: Higher local efficiency in the lateral OFC was positively associated with longer collimation time, while higher degree in the medial OFC was negatively associated. These results together suggest that both lateral and medial parts of the OFC contribute to the decision to wait before firing. The OFC is well known for its role in reward valuation and impulse inhibition, and these findings are consistent with that function.

Table 5.2 shows selected highlights from the nodal regression results.

Metric	Region	Outcome	β	p-value	q-value	Theme
local_eff	Temporal_Pole_Mid_L	colTimer_56%	0.0053	4.0e-16	4.8e-14	Temporal (affect)
degree	Cerebelum_Crus2_R	colTimer_56%	-0.000675	5.9e-09	7.0e-07	Cerebellar (timing)

Metric	Region	Outcome	β	p-value	q-value	Theme
degree	Cerebelum_Crus2_L	colTimer_56%	-0.000680	2.7e-08	1.6e-06	Cerebellar (timing)
local_eff	OFClat_L	colTimer_56%	0.0056	1.1e-07	6.9e-06	OFC (inhibition)
betweenness	Frontal_Inf_Oper_R	colTimer_54%	2.715	3.0e-06	3.6e-04	Frontal (confirmed)
degree	OFCmed_R	colTimer_56%	-0.000325	2.5e-04	2.1e-03	OFC (inhibition)
degree	Rectus_R	colTimer_56%	-0.000321	2.6e-04	2.1e-03	Medial frontal

Table 5.2. FDR-significant nodal regression highlights. *81 total FDR-significant findings from 2,520 models. local_eff = local efficiency. colTimer = collimationTimer threshold. Selected highlights from three main anatomical*

5.9 Multivariate and Machine Learning Analysis Results

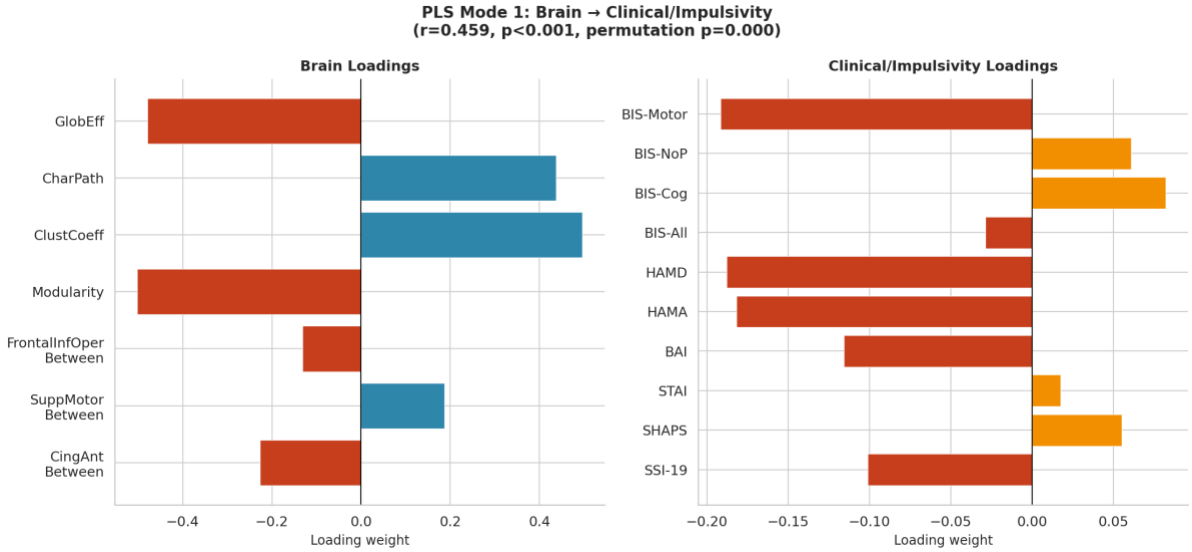


Figure 5.11 Multivariate Brain-Behavior Structure — PLS and Ridge Regression. *Left: PLS latent variable 1 — brain (graph metrics) versus clinical (scales) composite scores (r=0.459, permutation p<0.001, N=50 permutations). Right: Ridge LOO-CV predicted vs. actual SSI-19 scores (R^2=0.236, p<0.001). Both analyses demonstrate significant multivariate brain-clinical associations beyond the univariate findings.*

Partial Least Squares (PLS) analysis identified a significant first latent dimension (r = 0.459, permutation p < 0.001). The brain side of this dimension was mainly driven by global efficiency (negative loading), clustering coefficient (positive), and small-worldness sigma (positive). The clinical side was mainly driven by SSI-19, BIS-11 total, and HAMD. This means that a single dimension captures a meaningful shared structure between brain network organization and clinical severity in this sample.

Ridge regression with leave-one-out cross-validation predicted SSI-19 suicidal ideation scores from the seven confirmatory brain metrics with $R^2 = 0.236$ ($p < 0.001$). This indicates that approximately 24% of the variance in suicidal ideation severity can be predicted from structural brain connectivity alone, even after leave-one-out validation. For a neuroimaging prediction task with a relatively small sample, this is a clinically meaningful result.

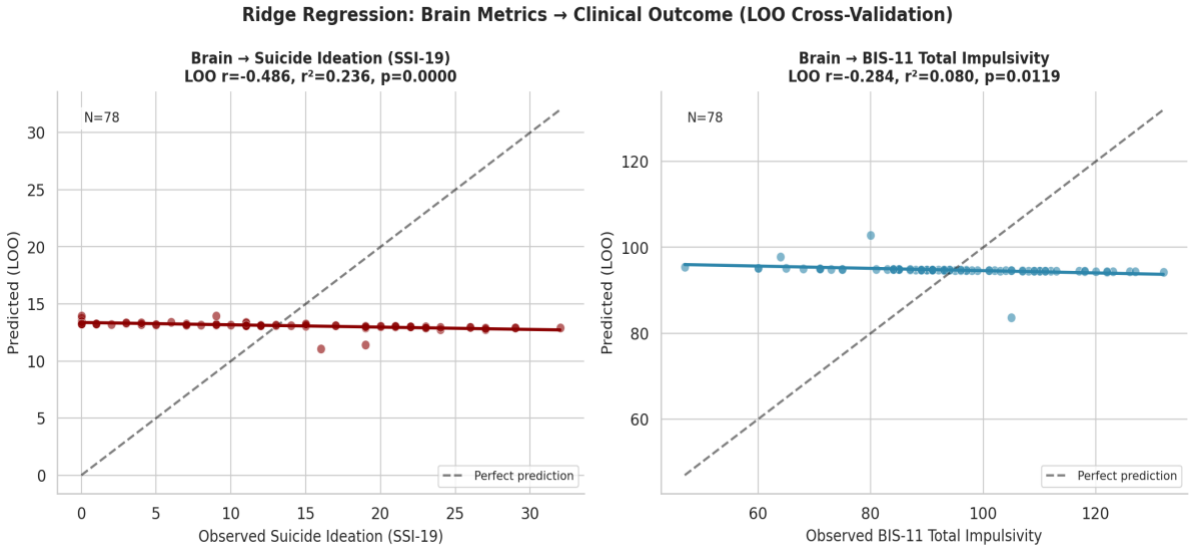


Figure 5.12 K-means Clustering — Brain Connectivity Patient Subgroups. *K-means clustering (k=2, silhouette=0.405) partitions patients into two subgroups based on connectome topology (all seven confirmatory metrics). Cluster A (lower global efficiency, higher modularity, lower hub betweenness) is associated with higher BIS-11 total and SSI-19 scores. Cluster B shows the reverse pattern. Group differences in clinical scales shown with error bars (mean ± SD).*

K-means clustering with k = 2 — chosen based on silhouette analysis (silhouette = 0.405) — identified two patient subgroups based on connectome topology. Cluster A

patients had lower global efficiency, higher modularity, and lower hub betweenness, and also showed higher BIS-11 total scores and higher SSI-19 suicidal ideation. Cluster B patients had more integrated networks and lower clinical severity. This unsupervised finding is consistent with the regression results — less integrated and more segregated brain networks are associated with higher impulsivity and greater suicide risk in this MDD sample.

5.10 Age Sensitivity Analysis

5.10.1 Main Sensitivity: Age as Covariate

All three FDR-significant brain–VR associations remained significant after age was added as a covariate in both model specifications: Model A, where age was entered as a continuous variable, and Model B, where age was entered as a binary age-group variable. The beta coefficients remained highly consistent across all models, indicating that the original findings were not explained by age-related differences in the sample.

Table 5.3 Age sensitivity analysis results for the three confirmed brain–VR findings. β = unstandardized regression coefficient; q = BH-FDR-corrected p-value; ΔR^2 = incremental R-squared representing the unique contribution of the brain metric.

Brain Metric × VR Outcome	β (Original)	β (+ Age, cts.)	β (+ Age grp.)	ΔR² (brain metric unique variance)
Betweenness_FIO_R × collimationTimer_54%	2.715 (q = 0.0016)	2.708 (q = 0.0018)	2.715 (q = 0.0018)	21.4–21.6%
Char. Path Length × collimationTimer_56%	0.0050 (q = 0.0081)	0.0050 (q = 0.0063)	0.0050 (q = 0.0056)	20.4–20.6%
Clustering Coeff. × collimationTimer_56%	24.10 (q = 0.0081)	27.10 (q = 0.0040)	25.48 (q = 0.0056)	20.8–22.1%

Table 5.3 *β = unstandardized regression coefficient; q = BH-FDR-corrected p-value; ΔR² = brain metric unique variance contribution, calculated as R²_{full} – R²_{baseline}. FIO_R = right inferior frontal opercularis. All three findings remained FDR-significant (q < 0.01) across all model specifications.*

5.10.2 Supplementary Age Analyses

We ran additional analyses to further examine the role of age in the brain–VR associations.

Age × brain metric interaction models. We found three FDR-significant interaction effects across the 518 models. Characteristic path length × collimationTimer_56% (β_interaction = −0.0202, q < 0.0001) and clustering coefficient × collimationTimer_56% (β_interaction = −70.40, q < 0.0001) showed significantly stronger effects in adolescents than in adults. In contrast, the right inferior frontal opercularis betweenness finding did not show significant age moderation (β_interaction = −0.067, p = 0.95), suggesting that this association was stable across both age groups.

Subgroup analyses. When we repeated the analyses separately within each age subgroup, all three original findings survived FDR correction in the adolescent subgroup (n = 41). In total, we found eight FDR-significant associations in adolescents, including five additional findings that were not detected in the full sample. In the adult subgroup (n = 37), no associations survived FDR correction. The effects were markedly weaker in adults, and in some cases the direction of the association was reversed. For example, clustering coefficient × collimationTimer_56% showed a strong positive association in adolescents ($\beta = 68.84$, $p < 0.0001$) but not in adults ($\beta = -1.63$, $p = 0.70$). These subgroup findings should be interpreted with caution because of the reduced sample size and lower statistical power in the adult group.

BIS-11 models with age covariate. When age was added to the 28 brain–BIS-11 models, no associations survived FDR correction (minimum $q = 0.155$). Beta coefficients changed by approximately 5–20% relative to the original models, suggesting that age shares partial overlap with both brain network properties and self-reported impulsivity scores.

Mediation analysis. Age did not significantly predict VR performance in any of the three mediation models (Path c: $p = 0.54–0.99$), and no significant indirect effects were observed, as all bootstrapped 95% confidence intervals crossed zero. In contrast, Path b — representing the association between brain metric and VR behavior — remained highly significant in all three models ($p < 0.0001$). Taken together, these results indicate that the confirmed brain–VR associations were direct and were not explained by age-mediated pathways.

5.11 White Matter Tract Analysis (JHU Atlas)

We tested the relationship between 48 JHU white matter tracts (FA, MD, RD, AD) and the two primary VR outcomes using partial Spearman correlation, controlling for head motion and gender (N = 74; four subjects did not have JHU data). Eight low-coverage brainstem tracts were excluded, resulting in 320 models. Benjamini–Hochberg FDR correction was applied within each metric–outcome group.

No associations remained significant after FDR correction. Eight nominal associations were observed at $p < 0.05$. The strongest result was axial diffusivity (AD) in the left fornix/stria terminalis, which showed a negative association with collimationTimer_56% ($\rho = -0.29$, $p = 0.012$, $q = 0.33$). Bilateral sagittal stratum AD ($\rho = -0.27$ and -0.26, $p < 0.025$) and left cingulum hippocampus FA ($\rho = +0.25$, $p = 0.030$) showed the next strongest associations. All eight findings were related to the 56% threshold outcome.

Previous studies have reported white matter microstructural abnormalities in depressed patients with suicidal behavior, especially reduced fractional anisotropy in tracts such as the corpus callosum, inferior fronto-occipital fasciculus, and anterior thalamic radiation. A recent meta-analysis of 14 DTI studies also reported reduced FA across frontal and limbic pathways in individuals with suicidal thoughts and behaviors. Our tract-level analysis did not replicate these findings. One likely reason is the limited sample size (N = 74), which reduces statistical power to detect tract-specific effects. Future studies with larger samples should examine whether specific white matter pathways are linked to both suicidal ideation and inhibitory control behavior.

Chapter 6: Discussion

6.1 Overview of Findings

In this thesis, we examined the relationship between structural brain connectivity and impulsive behavior in MDD patients across several levels of analysis — from global network properties to individual brain regions, modular organization, and fine-grained motor behavior during a VR task.

The main findings can be summarized as follows:

1. Three global brain metrics survived FDR correction — frontal hub betweenness, characteristic path length, and clustering coefficient — all of which predicted how long participants maintained their aim before firing. This supports the idea that frontoparietal network organization plays a role in deliberate motor control.
2. The expanded nodal analysis identified temporal pole local efficiency and cerebellar degree as the strongest regional predictors. These regions were not among the three pre-specified hubs, which suggests that a broader network — including areas involved in affective appraisal and internal timing — contributes to deliberate aiming behavior.
3. The within-module Z-score showed consistent positive associations with BIS-11 impulsivity subscales, suggesting that stronger provincial hub organization in the brain is linked to higher trait impulsivity in MDD.
4. The VR Cognitive Impulsivity composite (D2) predicted SSI-19 suicidal ideation scores ($q = 0.024$), showing that an objective behavioral measure from a VR task can capture clinically meaningful information about suicide risk.
5. The 0.6-second pre-shot window was empirically validated — the strongest individual clinical correlation occurred at exactly 0.6 seconds, and bootstrap analysis found no reason to use a shorter window for this population.
6. Multivariate analyses confirmed that the brain-clinical relationship is robust and has meaningful predictive value (PLS $r = 0.459$; Ridge $R^2 = 0.236$).

6.2 Frontal Hub Connectivity and Motor Control

The association between Frontal_Inf_Oper_R betweenness and collimationTimer_54% is probably the most theoretically grounded finding in this thesis. The frontal inferior opercularis is the structural counterpart of the inferior frontal gyrus (IFG), a region that has been consistently linked to response inhibition in tasks like stop-signal and go/no-go paradigms (Aron et al., 2004). Its betweenness centrality reflects how well it serves as a relay hub — connecting and passing information between other brain regions.

When this region has higher betweenness, patients tend to aim longer before firing. This is consistent with what response inhibition research would predict — a well-connected IFG hub is better able to slow down or stop a motor response when the conditions are not yet right. Although collimationTimer measures how long a participant keeps aiming, rather than inhibitory control directly, more deliberate aiming may reflect better ability to regulate responses during the task.

The path length and clustering findings add another layer to this picture. Higher characteristic path length — which reflects less global integration — and higher clustering coefficient — which reflects more local segregation — both predicted longer collimation time. This may seem surprising at first. Why would a less globally integrated network support better impulse control? One possible explanation is that in the context of fine motor deliberation, more modular networks may protect local motor circuits from being disrupted by activity elsewhere in the brain. This kind of "computational buffer" interpretation is consistent with ideas about focused attention, and it would be worth testing in future work.

The right-sided pattern of this finding is supported by previous work. Studies have shown that depression is often linked to changes in the right side of the brain, especially in prefrontal regions [29]. Based on this, the right IFG can be seen as an important region for inhibition and control. In our results, we also found that the right inferior frontal operculum, and not the left side, is related to deliberate motor behavior. This is

generally consistent with this idea, although testing hemispheric differences was not the main aim of this study.

Anxiety may also help explain this right-sided effect. One fMRI study showed that in patients with both depression and anxiety, brain activity shifts more to the right side, and the rIFG becomes more involved in controlling responses under threat [30]. In our sample, anxiety levels are relatively high (mean HAMA = 23.2, mean BAI = 22.9). This suggests that the rIFG finding may reflect a combined effect of depression and anxiety. Patients with a more intact rIFG network may be better able to engage this inhibitory system during the VR aiming task, although this is an indirect interpretation based on structural connectivity.

At the clinical level, the right IFG has also been linked to suicidal risk. One study reported reduced gray matter in the right inferior frontal region and altered connectivity in patients who attempted suicide compared to those who did not (Journal of Affective Disorders, 2020 [31]). This is somewhat related to our findings. We observed a small association between rIFG betweenness and SSI-19 ($\rho = 0.15$, $p = 0.19$), and a stronger association at the VR level, where trigblock was related to SSI-19 ($\rho = 0.37$, $p = 0.0009$).

Together, these studies suggest a possible interpretation. Depression may involve greater right-hemisphere involvement [29], the rIFG may act as an important region for anxiety-related inhibitory control [30], and changes in this region have been linked to suicidal risk in other studies [31]. We present this as a speculative interpretation rather than a conclusion, since these relationships were not directly tested in our data and would require further validation.

6.3 Temporal Pole and Cerebellar Connectivity

The nodal analysis revealed two anatomical themes that go beyond the global findings. The most striking result was the temporal pole local efficiency finding ($p = 4\times10^{-16}$), which was much stronger than anything found in the confirmatory analysis. The temporal pole is not typically considered an impulse control region — it is more

commonly linked to semantic memory, social cognition, and affective processing (Olson et al., 2007). The fact that its local efficiency predicted collimation time suggests that the aiming decision in the VR task is not purely a motor process. It may also involve an emotional assessment of consequences — something like "what happens if I fire before I am ready?" — mediated through the temporal pole.

This fits with emotional regulation theories of impulsivity, which suggest that impulsive behavior in mood disorders is partly driven by difficulty imagining the consequences of actions, not just by poor motor inhibition. Patients with more efficient temporal pole connectivity may be better at representing these consequences, which motivates them to wait longer before firing. This is a novel hypothesis that could be tested in future studies using approaches such as TMS or lesion mapping focused on the temporal pole.

The cerebellar findings — where higher degree in eight cerebellar lobules was associated with shorter collimation time — are consistent with the cognitive cerebellar hypothesis (Schmahmann & Sherman, 1998). The cognitive parts of the cerebellum, particularly Crus1 and Crus2, are connected to the prefrontal cortex and anterior cingulate cortex, and are involved in internal timing, error prediction, and motor planning. Patients with lower cerebellar structural connectivity may have weaker predictive motor control, leading them to fire before their internal timing system has reached a "ready" state. This is consistent with the pattern seen in more impulsive patients.

6.4 Hub Organization and Trait Impulsivity

The positive associations between within-module Z-score and BIS-11 impulsivity (Zi × BIS Total: $\rho = +0.34$; Zi × BIS Motor: $\rho = +0.32$) suggest that patients whose brain nodes are more strongly connected within their own lobe — rather than across lobes — tend to be more impulsive. These are called provincial hubs, and they are different from connector hubs, which link different parts of the brain together.

In a healthy brain, a balance between provincial and connector hubs supports both local processing and cross-network communication. In MDD patients with elevated Zi, it is

possible that pathways connecting the prefrontal cortex to subcortical regions have weakened, pushing the network into a more locally segregated state. This would reduce the frontal lobe's ability to regulate limbic and motor activity — which is one of the key mechanisms thought to underlie impulsive behavior in MDD. It would be worth examining in future studies whether Zi is higher in impulsive MDD patients compared to healthy controls or less impulsive MDD patients.

The skewness findings provide a complementary view. While Zi measures how strongly individual nodes connect within their own lobe, skewness of the degree and strength distributions reflects the overall network structure — how uneven the connectivity is across all 120 nodes. Higher skewness means the network has a more extreme hub-periphery pattern, with a small number of highly connected hubs and many weakly connected nodes.

The positive association between skewness and BIS-11 impulsivity ($\rho = 0.30$) suggests that this type of uneven network organization is related to higher impulsivity in MDD. Together with the Zi finding, these results suggest that both local and global hub dominance are linked to impulsive behavior in this sample.

6.5 Impulsivity at Different Time Scales in Pre-Shot Behavior

The within-between decomposition analysis looked at whether the associations between movement features and impulsivity operated at the level of individual shots, stable person-level differences, or both. The main finding was that all 14 estimated beta coefficients were positive for both the within-person and between-person components. This consistent direction across all models suggests a coherent pattern rather than a chance finding.

The between-person effects show that impulsivity is reflected in a person's overall movement style. Participants who scored higher on BIS-11 — especially the Motor subscale — showed higher levels of speed variability, angular velocity, and tremor on

average across their whole session. This suggests that impulsivity as a stable trait is linked to a more erratic sensorimotor style during the VR task.

The within-person effects add something different. Even after accounting for each participant's own average movement level, shots that were more erratic than that person's own norm were still associated with higher BIS-11 scores. Since BIS-11 is a person-level measure that does not change from shot to shot, this within-person effect suggests that more impulsive individuals not only move more erratically on average — their movement also varies more around their own average from shot to shot.

Together, these findings suggest that VR pre-shot kinematics capture impulsivity-related information at two levels at the same time — stable individual differences and moment-to-moment fluctuations. This has an interesting clinical implication: a tool sensitive to within-person kinematic variation could potentially be used to track changes in impulsivity over time, for example across a treatment program. These results are exploratory and should be replicated with proper correction for multiple comparisons before drawing stronger conclusions.

6.6 Time Window Analysis and Implications for VR Tasks

The window profile was surprisingly flat across the 0.2 to 0.8 second range for clinical outcomes — the largest difference in mean absolute correlation between any two windows was less than 0.003. At the same time, the strongest individual correlation for SSI-19 occurred at exactly 0.6 seconds. These two observations together provide a useful picture of the method.

On one hand, the flat profile means that pre-shot kinematics are consistent predictors of clinical state across the whole range tested. The associations do not depend on choosing exactly the right window, which is reassuring — it suggests the findings are not an artifact of an arbitrary choice. On the other hand, the 0.6-second window still

performed best for the primary clinical outcome of interest in this sample, which supports keeping it as the reference window.

Looking at individual features, jerk-based measures performed best at shorter windows (0.2 s), while slower features like speed and path length performed better at longer windows (0.6 to 0.8 s). This suggests that different aspects of the movement unfold at different times — impulsive ballistic movements happen early, while deliberate fine adjustments happen later. A future approach that combines features from multiple windows could potentially improve clinical prediction.

6.7 Possible Links Between Brain, VR Behavior, and Clinical Symptoms

6.7.1 VR Cognitive Impulsivity and Suicidal Ideation

The FDR-significant association between the D2 Cognitive Impulsivity composite and SSI-19 suicidal ideation ($\rho = +0.37$, $q = 0.024$) is probably the most clinically important finding in this thesis. The SSI-19 measures current suicidal ideation severity — not just past history or general risk. The positive correlation means that patients with higher current suicidal ideation showed worse performance on the VR aiming challenge — more erratic aim, faster and less controlled movements, and lower accuracy.

This suggests that the VR shooting task may have some potential as a behavioral screening tool for suicidal ideation severity in MDD. A patient with a high VR cognitive impulsivity score is more likely to also report active suicidal ideation. However, this finding would need to be replicated before any clinical use, and future work should examine sensitivity and specificity thresholds and compare the VR measure's predictive accuracy against existing self-report instruments.

6.7.2 Brain, VR, and Clinical Relationships

The clinical correlation results suggest a possible link between brain measures, VR behavior, and clinical symptoms. Some brain network measures, especially rIFG betweenness and global clustering, appear to be related to how participants behave in

the VR task. This VR behavior may also relate to clinical scales such as suicidal ideation and anxiety. However, these relationships should be interpreted with caution.

The trigblock condition appears to be an important part of this pattern. It is not a skill-based measure, but instead reflects what a participant does when an action cannot be completed. When the gun is blocked, continuing to press may indicate difficulty in stopping a motor response. This is similar to findings in the rIFG literature, where impaired inhibitory control affects behavior. In our results, trigblock showed a positive association with SSI-19 ($\rho = 0.37$, $q = 0.053$). This suggests that participants with higher suicidal ideation may show this behavior more often. However, since this result is close to the FDR threshold, it should be interpreted carefully.

The result between collimationTimer and STAI may reflect a different pattern. Participants with higher anxiety tended to take more time before firing. One possible explanation is that anxiety increases hesitation or over-monitoring before making a decision. Previous studies suggest that the right IFG may play a role in controlling actions under anxiety, but in our case this interpretation is still tentative. It is possible that longer collimation time reflects a delay in action, but this is not fully clear.

Overall, these findings may suggest two behavioral patterns: one where the participant has difficulty stopping an action (trigblock), and another where the participant takes more time before acting (collimationTimer), possibly reflecting hesitation. These interpretations are exploratory and require further validation in future studies.

6.8 Limitations

There are several limitations to this study that are important to acknowledge.
Cross-sectional design: This study cannot establish causal direction. It is not possible to say whether brain network differences cause impulsive behavior, or whether impulsive behavior shapes brain connectivity over time. Longitudinal studies that track both brain connectivity and VR performance across treatment would help address this.

No healthy control group: The patient data range from light severe symptoms. However, without a healthy comparison group as controls, it remains to be confirmed that the brain-behavior associations found here are exactly specific to MDD or more general. To address this possible issue, the BIS-11 threshold of 72 was used as an internal benchmark, yet group comparisons with healthy controls would surely strengthen the interpretation.

Medication heterogeneity: Participants were taking different types of medication, including antidepressants, anxiolytics, and mood stabilizers. These medications can affect both white matter structure and motor behavior. Medication type and dose were not included as covariates due to the sample size, but this is something future studies should address.

Lobe-based modular indices: The within-module Z-score and participation coefficient were calculated using anatomical lobe assignments rather than data-driven community detection. This means the modular metrics are approximations, and results should be interpreted with that in mind.

No replication sample: All findings come from a single site and a single dataset. Independent replication in a comparable MDD sample using the same VR protocol is needed before these results can be considered reliable enough for clinical use.

Influential observations: It was examined using Cook's Distance. The clustering coefficient result was sensitive to the most extreme observation (m041, Cook's D = 4.67) and should be interpreted with caution until it is replicated. In contrast, the rIFG betweenness and characteristic path length results were robust. Removing their most influential observations strengthened the associations, suggesting that these findings are not driven by outliers.

VR experience: Participants had different levels of VR gaming experience, which could affect motor accuracy in the task. Future studies should record VR experience as a covariate.

Statistical power: The study had 92% power to detect medium-sized effects (Cohen's $f^2 \geq 0.15$) at an uncorrected $\alpha = 0.05$ level. However, after applying BH-FDR correction across 518 models, the effective power for any single association dropped to around 26% at a medium effect size. This means that smaller effects may not have been detectable within this corrected framework. The absence of FDR-significant associations in some analyses should not be taken as evidence that no relationship exists — it may simply mean that any effect present is small and would require a larger sample or more targeted hypotheses to detect reliably.

White matter microstructure analysis: The analysis using 48 JHU tracts did not show any associations that survived FDR correction (N = 74). The detailed tract-level results are reported in Section 5.11 and are treated as supplementary findings.

6.9 Age Effects and Brain-Impulsivity Relationship in MDD

The age sensitivity and supplementary analyses suggest that the brain–impulsivity relationship in this sample may have an important developmental component. Although all three-primary brain–VR associations remained significant after adjusting for age, the subgroup analyses showed that these effects were much stronger in adolescents and were largely absent in adults. This pattern should be interpreted carefully.

One possible explanation is that white matter networks are still developing during adolescence. Because of this, individual differences in structural connectivity may have a stronger effect on behavior during this stage of development. In adolescents with MDD, variation in frontal hub connectivity, global path length, and clustering coefficient may therefore relate more directly to real-time motor inhibition behavior, as measured by the VR collimation timer. In adults, where white matter organization may be more

stable, the relationship between structural network topology and impulsive behavior may depend more on other influences — such as functional brain dynamics, illness history, or learned behavioral patterns — which were not captured in this cross-sectional DTI analysis.

At the same time, a statistical explanation must also be considered. The adult subgroup (n = 37) had lower statistical power than both the adolescent subgroup (n = 41) and the full sample (n = 78). The post-hoc power analysis showed that the full sample had 80% power to detect effects of approximately $f^2 \geq 0.104$. In the adult subgroup, the minimum detectable effect size increased to about $f^2 = 0.18$. This means that smaller true effects may not have been detected in adults. The lack of significant findings in the adult subgroup should therefore not be interpreted as definitive evidence that the relationship is absent. However, the interaction models — which directly tested age-related differences in the full sample — showed significant moderation for two of the three confirmed findings. This provides stronger support for the interpretation that the brain–VR relationship is genuinely stronger in adolescents, particularly for characteristic path length and clustering coefficient.

The mediation analysis provided additional clarification. Age did not show a significant indirect effect on VR behavior through any of the three confirmed brain metrics. In other words, age was not related to impulsive shooting behavior through these network measures. Instead, the findings suggest that the brain–VR associations are direct, while age influences the strength of these associations rather than explaining them away as a confound.

Overall, these results suggest that developmental stage may shape how structural brain organization relates to impulsive behavior in MDD. Future studies should examine this question more directly using age-matched subgroups, larger developmental samples, and models that explicitly test developmental trajectories, such as age × brain × behavior interactions.

Chapter 7: Conclusions and Future Directions

7.1 Summary

This thesis has shown that structural brain network organization in MDD is related to impulsive behavior measured through a VR shooting task, and that these relationships also extend to clinical outcomes including suicidal ideation. The confirmatory analysis found three FDR-significant associations between global brain metrics and VR task performance. The expanded nodal analysis identified 81 additional significant associations, with the temporal pole and cerebellum standing out as important regions that were not originally predicted. The within-module Z-score provided a useful brain-level marker of trait impulsivity. The VR Cognitive Impulsivity composite predicted SSI-19 suicidal ideation scores with FDR significance, which supports the clinical usefulness of the VR task. Finally, the 0.6-second pre-shot window was confirmed as the best window for predicting suicidal ideation in this sample.

7.2 Key Contributions

This thesis makes several contributions to the field:

1. This is one of the first studies to use VR shooting task kinematics as an objective behavioral measure of impulsivity in MDD, with a full analysis of different pre-shot time windows.
2. It presents one of the largest nodal-level connectomics analyses of VR behavior in an MDD sample — covering 120 brain regions, 480 nodal features, and 78 patients.
3. The 0.6-second pre-shot window was validated using bootstrap methodology rather than simply being adopted from the athlete shooting literature, which gives it stronger empirical support.
4. Temporal pole local efficiency and cerebellar degree were identified as new structural brain correlates of deliberate motor control — regions that had not been previously linked to VR-based impulsivity measures.

5. Within-module Z-score was shown to capture BIS-11 impulsivity variance independently of global network metrics, adding a new modular-level marker to the existing literature.
6. The VR Cognitive Impulsivity composite showed FDR-significant convergence with SSI-19 suicidal ideation scores, supporting the potential of this VR task as a clinical screening tool.

7.3 Future Directions

Several directions could build on the findings of this thesis:

Longitudinal VR tracking: Future studies could reassess VR behavior and brain connectivity after treatment to test whether improvements in network organization predict reductions in impulsivity. This would help establish whether the brain-behavior relationships found here are stable traits or state-dependent markers that change with treatment.

Full community detection: The modular metrics in this thesis — particularly Zi and Pi — were calculated using anatomical lobe assignments rather than data-driven methods. Future work could use Louvain community detection applied to individual connectivity matrices to get more precise modular indices, which may reveal stronger or more specific associations.

Finer window analysis: The pre-shot window analysis in this thesis tested windows in steps of 0.1 seconds. A finer resolution analysis — for example using 0.05-second steps between 0.40 and 0.80 seconds — could help determine whether the SSI-19 optimum falls precisely at 0.6 seconds or is slightly shifted in one direction.

Temporal pole targeted analysis: Given the surprisingly strong finding involving the left middle temporal pole, a targeted analysis using seed-based tractography from this region could help identify the specific white matter pathways that drive this

association. This would give a clearer picture of how the temporal pole connects to the broader deliberation circuit.

Multi-window feature engineering: The window analysis showed that jerk and acceleration features perform best at shorter windows (0.2 s), while speed and path length features perform better at longer windows (0.8 s). Combining features from both short and long windows into a single composite kinematic score could improve prediction of SSI-19 in future studies.

Clinical translation: Future work should establish the sensitivity and specificity of the D2 VR composite and peak hand acceleration at 0.6 seconds for detecting active suicidal ideation. A receiver operating characteristic (ROC) analysis using SSI-19 ≥ 10 as a threshold could provide practically useful cutoff values for clinical screening.

Age-stratified and developmental analyses: Our findings suggest that the brain–VR associations may be stronger in adolescent patients with MDD than in adult patients. Future studies should recruit age-stratified samples with sufficient power within each developmental group, with a recommended sample size of at least 60 participants per group. These studies should also include age × brain × VR three-way interaction models to test whether developmental stage changes the relationship between structural brain connectivity and impulsive behavior. Longitudinal follow-up would be especially valuable to determine whether this brain–impulsivity link changes over time as patients age and as white matter maturation continues.

References

[1] World Health Organization. (2023). Depressive disorder (depression). WHO Fact Sheet. https://www.who.int/news-room/fact-sheets/detail/depression

[2] Moeller, F. G., Barratt, E. S., Dougherty, D. M., Schmitz, J. M., & Swann, A. C. (2001). Psychiatric aspects of impulsivity. American Journal of Psychiatry, 158(11), 1783–1793.

[3] Patton, J. H., Stanford, M. S., & Barratt, E. S. (1995). Factor structure of the Barratt Impulsiveness Scale. Journal of Clinical Psychology, 51(6), 768–774.

[4] Swann, A. C., Lijffijt, M., Lane, S. D., Steinberg, J. L., & Moeller, F. G. (2009). Severity of bipolar disorder and impulsivity: a neurobiological framework. Bipolar Disorders, 11(5), 503–512.

[5] Dolan, S. L., Bechara, A., & Nathan, P. E. (2008). Executive dysfunction as a risk marker for substance abuse: the role of impulsive personality traits. Behavioural Sciences & the Law, 26(6), 799–822.

[6] Parsons, T. D., & Rizzo, A. A. (2008). Affective outcomes of virtual reality exposure therapy for anxiety and specific phobias: a meta-analysis. Journal of Behavior Therapy and Experimental Psychiatry, 39(3), 250–261.

[7] Korgaonkar, M. S., Erlinger, M., Brammer, M., Williams, L. M., & Grieve, S. M. (2020). Widespread structural brain changes in major depressive disorder: evidence from mega-analysis. NeuroImage: Clinical, 27, 102250.

[8] Bullmore, E., & Sporns, O. (2009). Complex brain networks: graph theoretical analysis of structural and functional systems. Nature Reviews Neuroscience, 10(3), 186–198.

[9] Rubinov, M., & Sporns, O. (2010). Complex network measures of brain connectivity use and interpretations. NeuroImage, 52(3), 1059–1069. https://pubmed.ncbi.nlm.nih.gov/19819337/

[10] Mulders, P. C., van Eijndhoven, P. F., Schene, A. H., Beckmann, C. F., & Tendolkar, I. (2015). Resting-state functional connectivity in major depressive disorder: a review. Neuroscience & Biobehavioral Reviews, 56, 330–344.

[11] Smith, R. E., Tournier, J. D., Calamante, F., & Connelly, A. (2015). SIFT2: Enabling dense quantitative assessment of brain white matter connectivity using streamlined tractography. NeuroImage, 119, 338–351. https://pubmed.ncbi.nlm.nih.gov/26163802/
[12] Yeh, C. H., Smith, R. E., Liang, X., Calamante, F., & Connelly, A. (2016). Connectomes from streamlines tractography: assigning streamlines to brain parcellations is not trivial but highly consequential. NeuroImage, 199, 142–158. https://pubmed.ncbi.nlm.nih.gov/31082471/
[13] Tournier, J. D., et al. (2022). Quantitative streamlines tractography methods and inter-subject normalisation. Aperture Neuro. https://apertureneuro.org/article/81056
[14] Civier, O., et al. (2024). Structural connectome construction using constrained spherical deconvolution in multi-shell diffusion-weighted MRI. Nature Protocols. https://www.nature.com/articles/s41596-024-01129-1
[15] Andersson, J. L. R., & Sotiropoulos, S. N. (2016). FSL Eddy — diffusion MRI motion correction. FSL Documentation. https://fsl.fmrib.ox.ac.uk/fsl/docs/diffusion/eddy/users_guide/index.html
[16] Yendiki, A., Koldewyn, K., Kakunoori, S., Kanwisher, N., & Fischl, B. (2014). Spurious group differences due to head motion in a diffusion MRI study. NeuroImage, 88, 79–90. https://pubmed.ncbi.nlm.nih.gov/24269273/
[17] Zalesky, A., Fornito, A., & Bullmore, E. T. (2010). Network-based statistic: identifying differences in brain networks. NeuroImage, 53(4), 1197–1207. https://pubmed.ncbi.nlm.nih.gov/20600983/
[18] Winkler, A. M., Ridgway, G. R., Webster, M. A., Smith, S. M., & Nichols, T. E. (2014). Permutation inference for the general linear model. NeuroImage, 92, 381–397. https://pubmed.ncbi.nlm.nih.gov/24530839/
[19] FSL. (n.d.). FDR correction — FSL statistics documentation. https://fsl.fmrib.ox.ac.uk/fsl/docs/statistics/fdr.html
[20] MRtrix3. (n.d.). tcksift2 command reference. https://userdocs.mrtrix.org/en/latest/reference/commands/tcksift2.html
[21] MRtrix3. (n.d.). tck2connectome command reference. https://userdocs.mrtrix.org/en/latest/reference/commands/tck2connectome.html
[22] Qin, J., et al. (2022). Adaptive multimodal neuroimage integration for major depression disorder detection. *Frontiers in Neuroinformatics*, 16, 856175.

https://www.frontiersin.org/journals/neuroinformatics/articles/10.3389/fninf.2022.856175/full
[23] Librenza-Garcia, D., et al. (2019). Machine learning in major depression: from classification to treatment outcome prediction. *CNS & Neurological Disorders — Drug Targets*, 17(7), 482–494. https://pmc.ncbi.nlm.nih.gov/articles/PMC6324186/
[24] Jiang, X., et al. (2025). Advancing early detection of major depressive disorder using multisite functional magnetic resonance imaging data: comparative analysis of AI models. *JMIR Mental Health*. https://pmc.ncbi.nlm.nih.gov/articles/PMC12282933/
[25] Zhao, Y., et al. (2025). Multi-atlas ensemble graph neural network model for major depressive disorder detection using functional MRI data. *Frontiers in Computational Neuroscience*, 19, 1537284. https://www.frontiersin.org/journals/computational-neuroscience/articles/10.3389/fncom.2025.1537284/full
[26] Huang, Y., et al. (2025). A virtual reality-based multimodal framework for adolescent depression screening using machine learning. *Frontiers in Psychiatry*, 16, 1655554.
https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2025.1655554/full
[27] Pollak, Y., et al. (2024). Virtual reality-assisted prediction of adult ADHD based on eye tracking, EEG, actigraphy and behavioral indices: a machine learning analysis of independent training and test samples. *Translational Psychiatry*, 14, 87.
https://www.nature.com/articles/s41398-024-03217-y
[28] Janssen, R. J., et al. (2023). Machine learning and brain imaging for psychiatric disorders. In *Machine Learning for Brain Disorders*. NCBI Bookshelf.
https://www.ncbi.nlm.nih.gov/books/NBK597503/
[29] Right brain, left brain in depressive disorders. Neuroscience & Biobehavioral Reviews, 2017, 78:178–191. doi: 10.1016/j.neubiorev.2017.04.021
[30] Co-occurring Anxiety Influences Patterns of Brain Activity in Depression. Cognitive, Affective, & Behavioral Neuroscience, 2010, 10(1):141–156. doi: 10.3758/CABN.10.1.141
[31] Atrophy of right inferior frontal orbital gyrus and frontoparietal functional connectivity abnormality in depressed suicide attempters. Journal of Affective Disorders, 2020.

[32] Zhang H, Li H, Yin L, Chen Z, Wu B, Huang X, Jia Z, Gong Q. Aberrant white matter microstructure in depressed patients with suicidality. *Journal of Magnetic Resonance Imaging*. 2022;55(4):1141–1150. doi:10.1002/jmri.27927
[33] Lim L, Radua J, Beyh A. Structural connectivity abnormalities in suicidal thoughts and behaviours: a meta-analysis. *Translational Psychiatry*. 2025;15:480. doi:10.1038/s41398-025-03699-4
[34] Karger, A., *Gender Differences in Depression.* Bundesgesundheitsblatt - Gesundheitsforschung - Gesundheitsschutz, 2014. **57**(9): p. 1092-1098.
[35] Bertakis, K.D., L.J. Helms, E.J. Callahan, R. Azari, P. Leigh, and J.A. Robbins, *Patient Gender Differences in the Diagnosis of Depression in Primary Care*. J Womens Health Gend Based Med, 2001. **10**(7): p. 689-698.
[36] Piccinelli, M. and G. Wilkinson, *Gender Difference in Depreession. Critical Review*. Br J Psychiatry, 2000. **177**: p. 486-492.
[37] Schuch, J.J., A.M. Roest, W.A. Nolen, B.W. Penninx, and P. de Jonge, *Gender differences in major depressive disorder: results from the Netherlands study of depression and anxiety*. J Affect Disord, 2013. **156**: p. 156-163.
[38] Essau, C.A., P.M. Lewinsohn, J.R. Seeley, and S. Sasagawa, *Gender Differences in the Developmental Course of Depression*. J Affect Disord, 2010. **127**: p. 185-190.
[39] Baca A, Kornfeind P. Stability analysis of motion patterns in biathlon shooting. *Human Movement Science*. 2012;31(2):295–302. DOI: 10.1016/j.humov.2010.05.008
[40] Köykkä M, Ihalainen S, Linnamo V, Ruotsalainen K, Häkkinen K, Laaksonen MS. Aiming strategy affects performance-related factors in biathlon standing shooting. *Scandinavian Journal of Medicine & Science in Sports*. 2021;31(3):573–585. DOI: 10.1111/sms.13864
[41] Congdon, E., Mumford, J. A., Cohen, J. R., Galvan, A., Canli, T., & Poldrack, R. A. (2012). Measurement and reliability of response inhibition. *Frontiers in Psychology*, 3, 37. https://doi.org/10.3389/fpsyg.2012.00037
[42] Borgnis, F., Baglio, F., Pedroli, E., Rossetto, F., Uccellatore, L., Oliveira, J. A. G., Riva, G., & Cipresso, P. (2022). Available virtual reality-based tools for executive functions: A systematic review. *Frontiers in Psychology*, 13, 833136. https://doi.org/10.3389/fpsyg.2022.833136

[43] Ajana, K., Everard, G., Lejeune, T., & Edwards, M. G. (2023). A feature and conjunction visual search immersive virtual reality serious game for measuring spatial and distractor inhibition attention using response time and action kinematics. *Journal of Clinical and Experimental Neuropsychology*, 45(3), 292–303. https://doi.org/10.1080/13803395.2023.2228669

[44] Dina, L. M., Pinti, P., & Smith, T. J. (2025). Measuring neurodevelopment of inhibitory control in children using naturalistic virtual reality. *Scientific Reports*, 15(1), 26944. https://doi.org/10.1038/s41598-025-11234-5

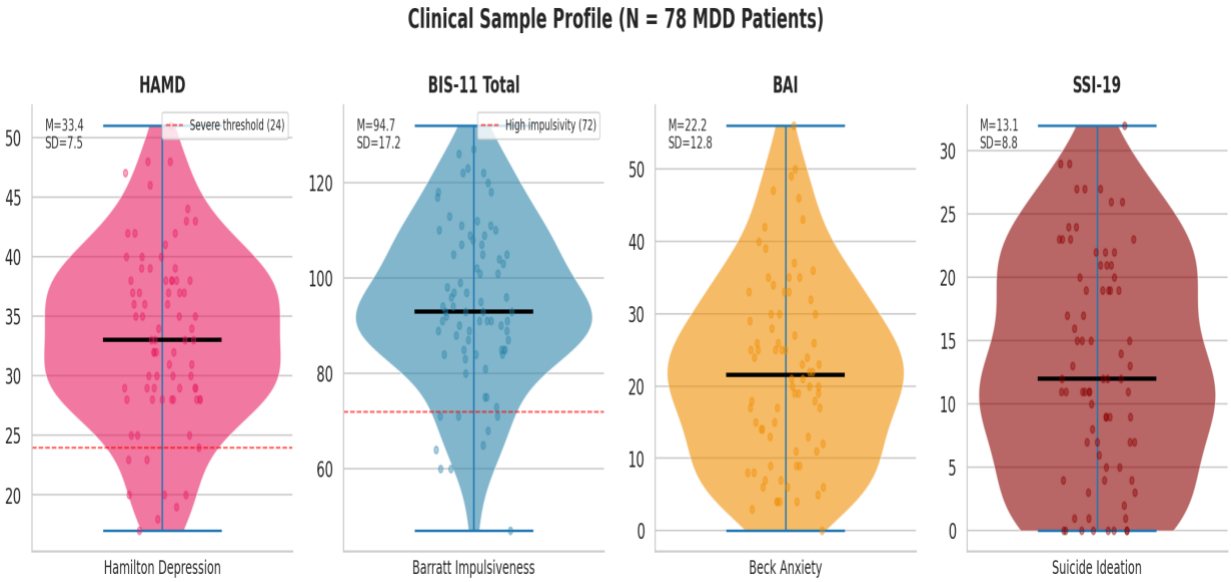
[45] Logan, G. D., Van Zandt, T., Verbruggen, F., & Wagenmakers, E. J. (2014). On the ability to inhibit thought and action: General and special theories of an act of control. *Psychological Review*, 121(1), 66–95. https://doi.org/10.1037/a0035230

[46] Dalley, J. W., Everitt, B. J., & Robbins, T. W. (2011). Impulsivity, compulsivity, and top-down cognitive control. *Neuron*, 69(4), 680–694. https://doi.org/10.1016/j.neuron.2011.01.020

Appendix A: Full Figure Gallery

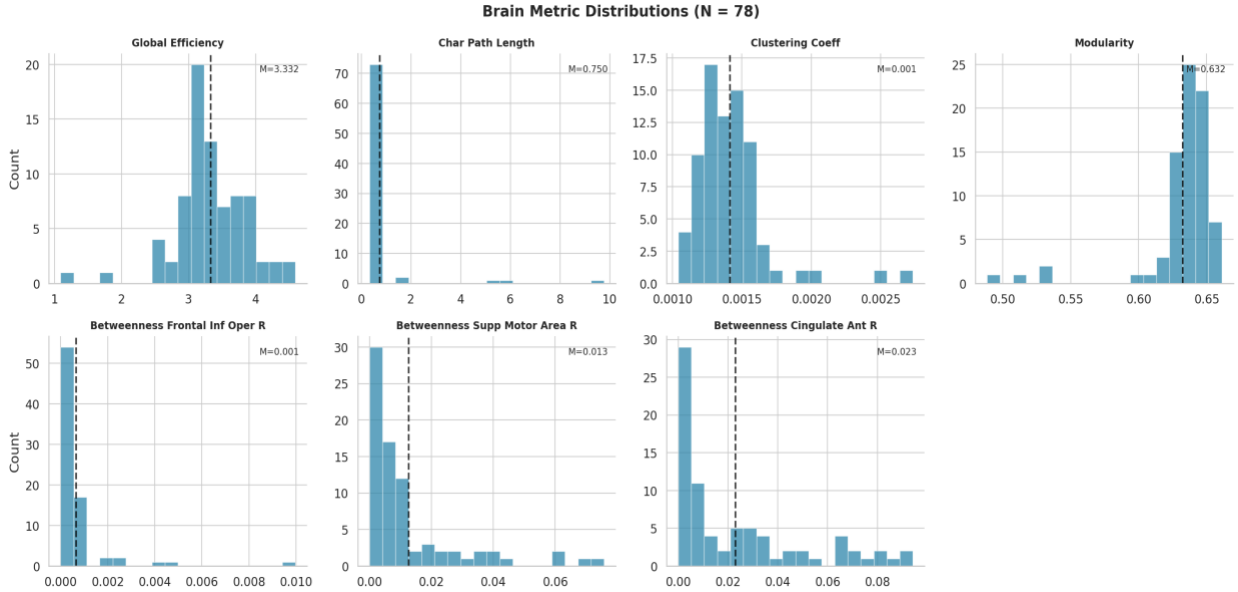
Original Confirmatory Figures

Figure A1. Study Design Overview.



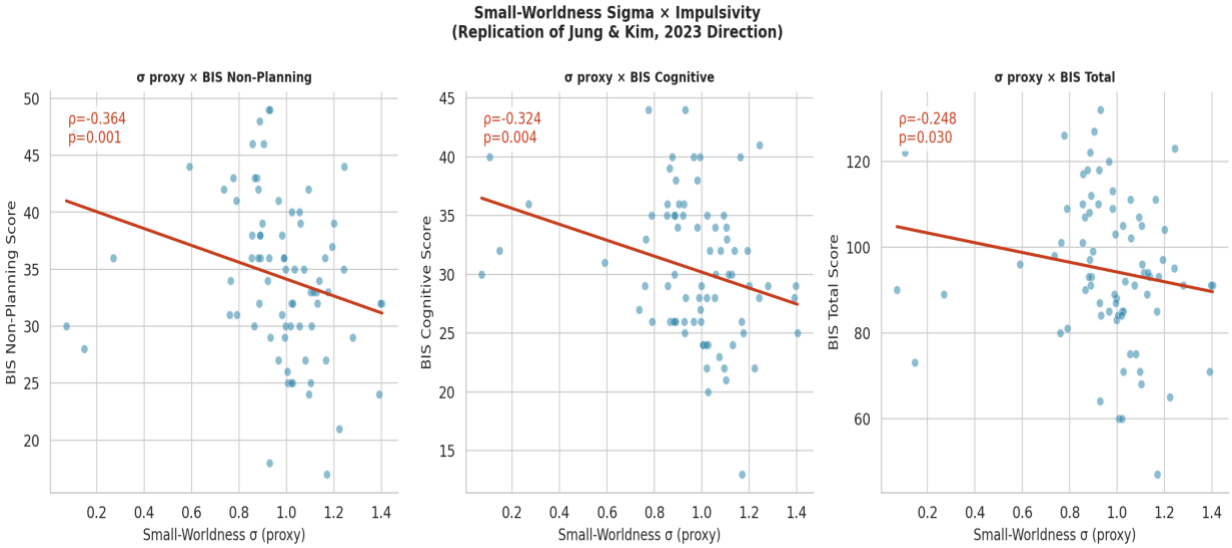
End-to-end pipeline from DTI/VR data acquisition through confirmatory analysis to machine learning outcomes.

Figure A2. Clinical Characteristics.



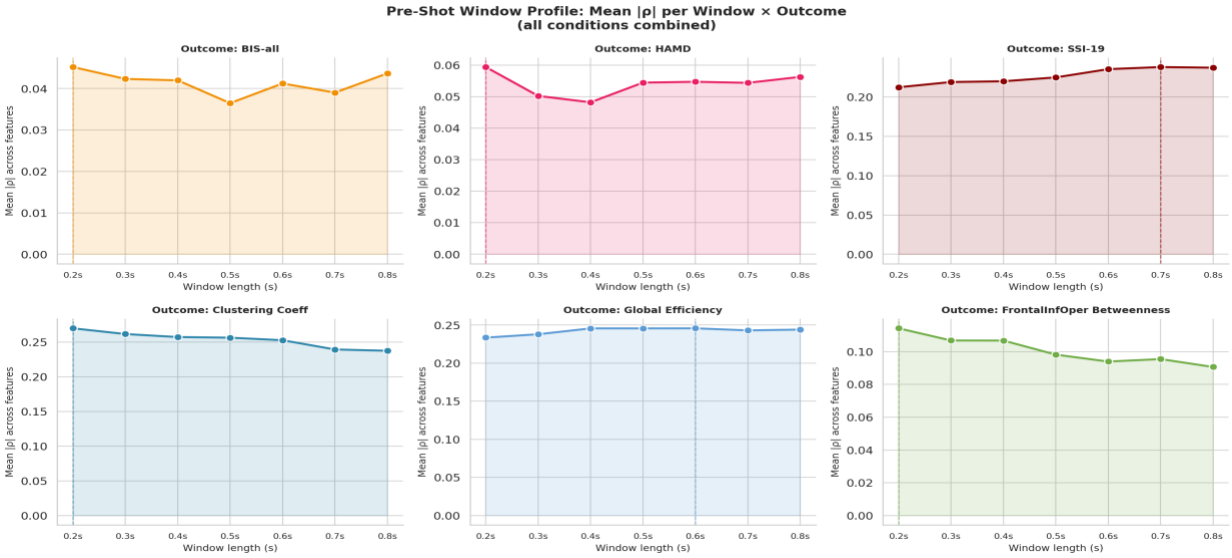
Histograms of key clinical variables. All BIS-11 values above the high-impulsivity threshold (72); HAMD mean=32.5 (severe depression).

Figure A3. Small-Worldness Analysis.



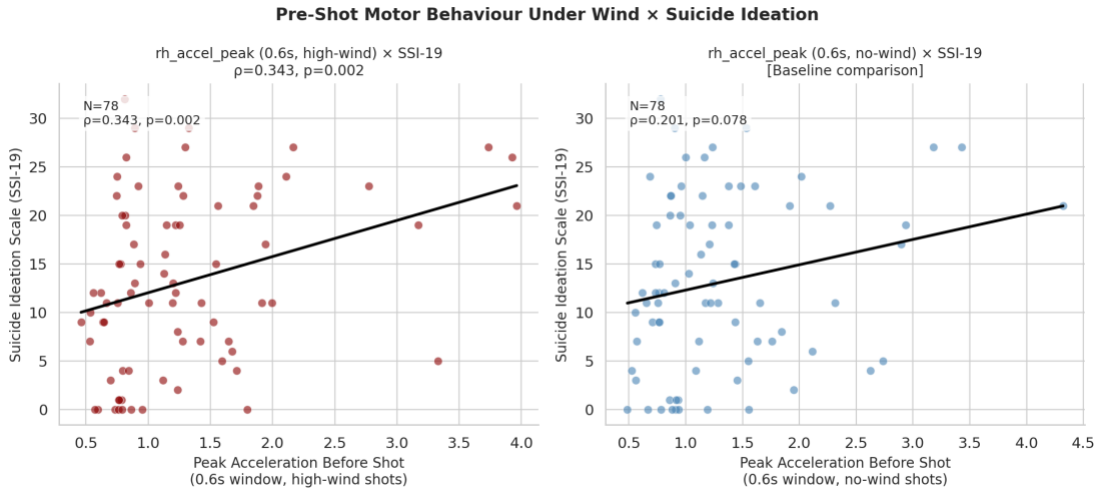
Sigma proxy distribution (left), gamma vs lambda scatter (center), and correlation with BIS-11 (right). Sigma negatively correlates with BIS-11 Non-Planning.

Figure A4. VR Composites Detailed



PCA loadings for D1 Motor, D2 Cognitive, and D3 Non-Planning composites.

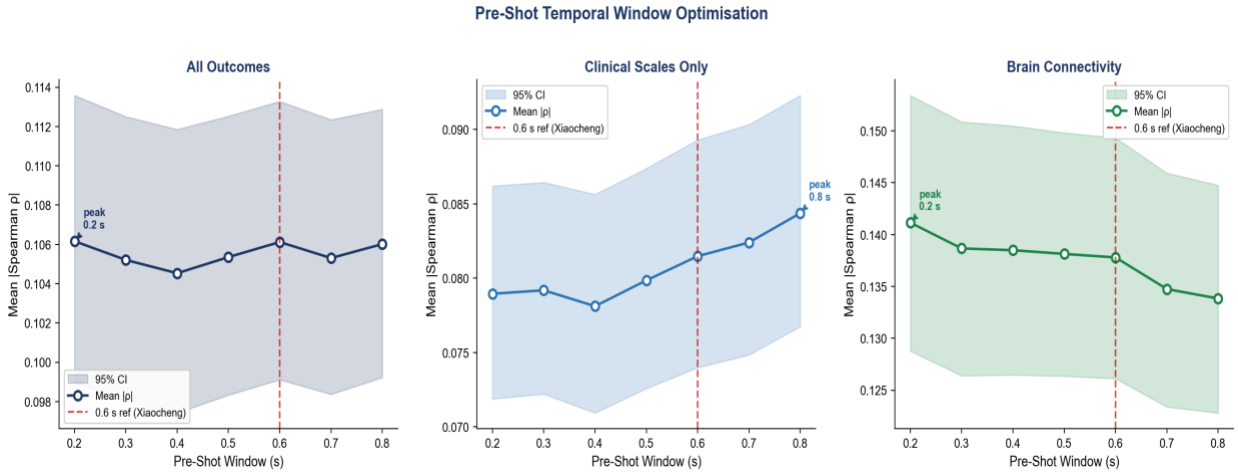
Figure A5. Pre-Shot Correlation Summary.



Overview of all 3,584-pre-shot feature × outcome correlations; nominal signals by window and condition.

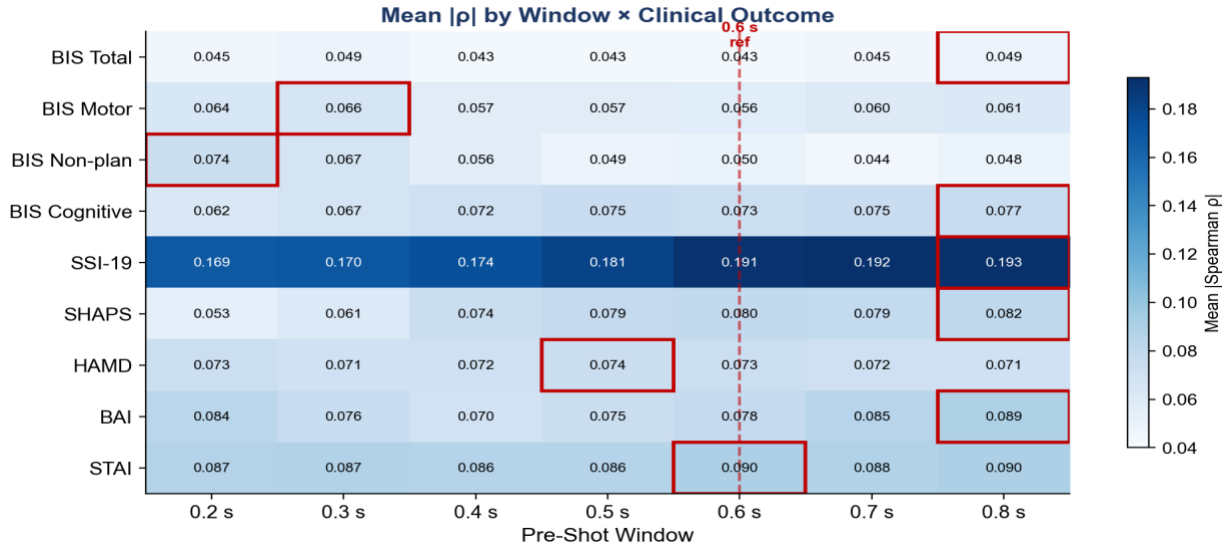
Extended Analysis Figures

Figure A7. Window Optimization Profiles



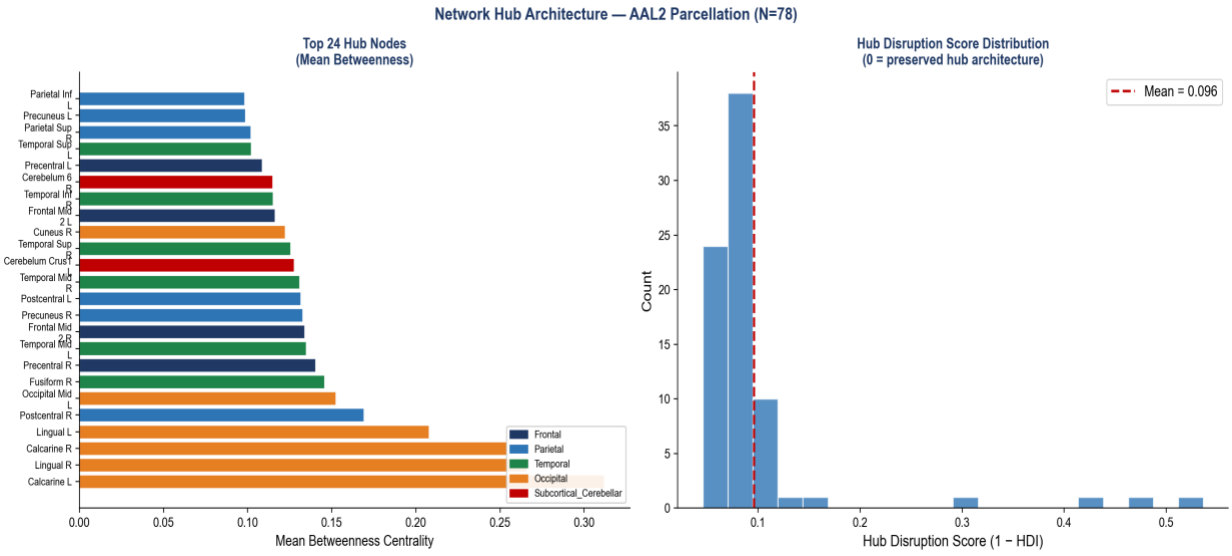
Three-panel window profile: all outcomes, clinical only, brain connectivity only.

Figure A8. Window × Outcome Heatmap



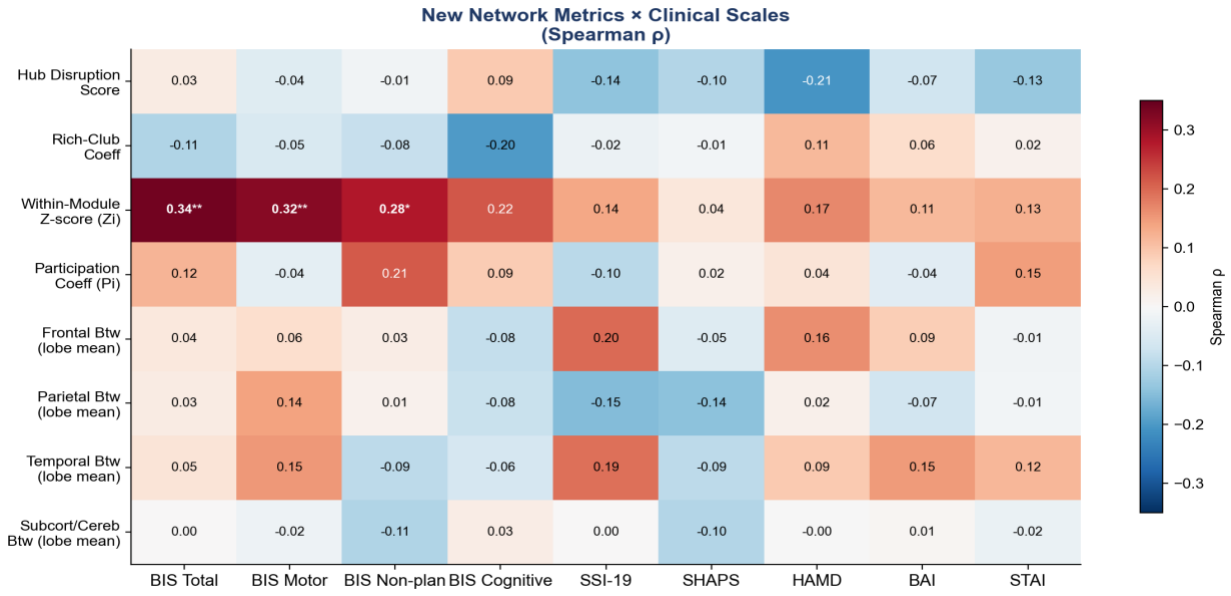
x Mean |ρ| heat map showing outcome-specific optimal windows.

Figure A9. Hub Architecture



Top 24 hub nodes by betweenness and hub disruption score distribution.

Figure A10. Additional Metrics × Clinical Heatmap



Eight additional modular network metrics against nine clinical scales.

Appendix B: Supplementary Tables

B1 Top 24 Hub Nodes

Rank	Region	Betweenness	Degree	Strength	Lobe
1	Calcarine_L	0.312	97.4	183.1	Occipital
2	Lingual_R	0.266	102.5	147.1	Occipital
3	Calcarine_R	0.254	94.0	150.5	Occipital
4	Lingual_L	0.208	101.4	151.9	Occipital
5	Postcentral_R	0.169	102.8	169.0	Parietal
6	Occipital_Mid_L	0.152	102.1	201.7	Occipital
7	Fusiform_R	0.145	93.7	121.7	Temporal

Rank	Region	Betweenness	Degree	Strength	Lobe
8	Precentral_R	0.140	100.9	146.1	Frontal
9	Temporal_Mid_L	0.135	104.7	280.2	Temporal
10	Frontal_Mid_2_R	0.134	105.0	174.8	Frontal
11	Postcentral_L	0.132	106.1	190.6	Parietal
12	Precentral_L	0.124	100.2	151.1	Frontal
13–24	… (remaining 12 hubs)	—	—	—	Mixed

Table B1. *Full top-24 hub node list*

B2 Bootstrap Window Optimization Results

Outcome Set	Bootstrap Mode	95% CI	P(opt≠0.6s)	P(opt<0.6s)	P(opt>0.6s)
All outcomes	0.2 s	[0.20, 0.80]	0.820	0.539	0.281
Clinical only	0.8 s	[0.20, 0.80]	0.852	0.201	0.650
Brain connectivity	0.2 s	[0.20, 0.80]	0.874	0.793	0.081

Table B2. *Bootstrap optimal window test. N=10,000 bootstrap iterations. Wide CIs reflect the flat profile and limited statistical power to distinguish adjacent windows.*

B3 Outcome-Specific Optimal Windows

Outcome	Optimal Window	Mean |ρ|	Top Feature	Top ρ
SSI-19	0.8 s	0.193	rh_accel_peak (hi-wind)	0.327
STAI	0.6 s	0.090	rh_jerk_mean	0.107
SHAPS	0.8 s	0.082	rh_accel_peak	0.229
HAMD	0.5 s	0.074	rh_speed_std	0.136
BAI	0.8 s	0.089	rh_speed_std	0.174
BIS Total	0.8 s	0.049	rh_jerk_mean	0.135
BIS Motor	0.3 s	0.066	rh_accel_peak	0.135
BIS Non-plan	0.2 s	0.074	rh_accel_peak	0.128
BIS Cognitive	0.8 s	0.077	rh_jerk_mean	0.148

***Table B3**. Outcome-specific pre-shot optimal windows. Most clinical outcomes optimize at 0.6–0.8 s; BIS Motor/Non-planning optimize earlier (0.2–0.3 s), suggesting different temporal dynamics for affective vs. trait impulsivity measures.*